Congruences and Divisibility for Fourth-Power Binomial Polynomials at 36

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October 16, 2026

Abstract

Let $S_n(t)=\sum_{j=0}^{n}\binom{n}{j}^4 t^j$. We prove the three assertions of Sun's conjecture A39(i): quadratic-form evaluations modulo p^2 of $\sum_{n<p} S_n(36)$, a weighted congruence for every odd prime, and divisibility by n of $\sum_{k<n}(8k+7)S_k(36)$ for every positive integer. Finite reflection identities connect these sums with fourth Franel periods. Modular coordinates, canonical Frobenius, and an integral specialization on a degree-three correspondence establish the congruences at both ordinary and supersingular primes. The divisibility argument uses a Newton expansion, ordinary connection congruences, and formal Brauer groups, with a separate treatment of the singular fiber at 5. The finite algebraic certificates used in the proof are included as ancillary files.

Keywords: supercongruences; Franel numbers; complex multiplication; K3 surfaces; formal Brauer groups.

1 Main results and conventions

For every integer $n\ge 0$, put

$$S_n(t)=\sum_{j=0}^{n}\binom{n}{j}^4 t^j,\qquad U_p(t)=\sum_{n=0}^{p-1}S_n(t),\qquad W_n(t)=\sum_{k=0}^{n-1}(8k+7)S_k(t).$$

A symbol $\left(\frac{a}{b}\right)$ with positive odd denominator denotes the Jacobi symbol, including value zero when a,b are not coprime. For odd prime denominator it is the Legendre symbol.

Theorem 1.1 (Quadratic-form congruences)**.** *Let $p>5$ be prime and set $\eta_p=((2/p),(p/3),(p/5))$. Then*

$$U_p(36)\equiv\begin{cases}4r^2-2p, & \eta_p=(1,1,1), \quad p=r^2+30s^2,\\ 12r^2-2p, & \eta_p=(-1,1,-1), \quad p=3r^2+10s^2,\\ 8r^2-2p, & \eta_p=(1,-1,-1), \quad p=2r^2+15s^2,\\ 2p-20r^2, & \eta_p=(-1,-1,1), \quad p=5r^2+6s^2,\\ 0, & \left(\frac{-30}{p}\right)=-1\end{cases}\pmod{p^2}. \tag{1.1}$$

In each of the first four cases the indicated representation exists with $r,s\in\mathbb{Z}$.

Theorem 1.2 (Weighted congruence)**.** *For every odd prime p, including $p=3,5$,*

$$W_p(36)\equiv p\left(\frac{p}{15}\right)\left(3+4\left(\frac{-6}{p}\right)\right)\pmod{p^2}. \tag{1.2}$$

Theorem 1.3 (Divisibility)**.** *For every positive integer n,*

$$\frac{W_n(36)}{n} \in \mathbb{Z}. \tag{1.3}$$

Part I proves theorems 1.1 and 1.2. Part II proves theorem 1.3, treating all primes, exponents, and cofactors. Within either part an unqualified numbered section refers to that part. The mnemonic names DUAL, TATE, CAN, and so on are retained in parentheses to make dependencies easy to track; the mathematical statements and equations are numbered automatically.

Part I
Congruences

I.1 Notation and dependencies

Throughout the proof use

$$H_0 = 0, \quad H_n = \sum_{i=1}^{n} \frac{1}{i}, \quad f_n = \sum_{j=0}^{n} \binom{n}{j}^4, \quad d_n = \sum_{j=0}^{n} \binom{n}{j}^4 (H_n - H_j), \quad \ell_n = 4d_n,$$

$$G(x) = \sum_{n\geq 0} f_n x^n, \quad L(x) = \sum_{n\geq 0} \ell_n x^n, \quad \theta = x\frac{d}{dx}, \quad \Delta(x) = (1-16x)(1+4x),$$

$$B = B_p(x) = \sum_{n=0}^{p-1} f_n x^n, \quad T = T_p(x) = \sum_{n=0}^{p-1} \ell_n x^n, \quad R = (1+4\theta)B, \quad a = \frac{1}{36}.$$

In expressions over $\mathbb{F}_p$, the coefficients of B, T are understood to be reduced modulo p. In particular all degree bounds on B in the pole comparison refer to its reduction. The formal modular coordinate is $q(x) = x\exp(L/G)$. To avoid confusing this inverse function with a Fourier variable, write $Q = e^{2\pi i\tau}$ when discussing eta products. The modular function itself is denoted $x(\tau)$. A local variable such as u or a cover coordinate z will always be declared explicitly.

The argument proceeds through finite reflection (DUAL), the differential and modular description of the Franel period (PFLOG and TATE), a finite two-digit calculation and an integral modular pole bound (DIGIT and POLES), and their global canonical-lift consequence (CAN). The zero-lifting lemma SSLIFT and the exact CM normalization CMPOINT separate the inert and split primes. CMRATIO and SPLIT prove the ordinary values; IRR proves the inert weighted value. DOUGALL then returns these Franel values to the original weighted sum.

I.2 Finite first-order reflection

Lemma I.2.1 (DUAL)**.** *Let p be an odd prime and $m \in \mathbb{Z}_p^\times$. Rational congruences are interpreted in $\mathbb{Z}_{(p)}$. Use the harmonic numbers and Franel numbers defined above, and put*

$$B_p(m^{-1}) = \sum_{k=0}^{p-1} m^{-k} f_k, \qquad D_p(m^{-1}) = \sum_{k=0}^{p-1} m^{-k} \sum_{r=0}^{k} \binom{k}{r}^4 (H_k - H_r).$$

Then

$$\boxed{\sum_{n=0}^{p-1} S_n(m) \equiv m^{p-1}(B_p(m^{-1}) - 4pD_p(m^{-1})) \pmod{p^2}.} \tag{I.2.1}$$

Proof. Interchange n, j, put $j = p-1-k$ and $n = j+\ell$. This gives the exact identity

$$\sum_{n<p} S_n(m) = m^{p-1} \sum_{k<p} m^{-k} \sum_{l=0}^{k} \binom{p-1-k+l}{l}^4.$$

For $0 \le l \le k < p$,

$$\binom{p-1-k+l}{l} = (-1)^l \binom{k}{l} \prod_{i=k-l+1}^{k} (1-p/i) \equiv (-1)^l \binom{k}{l}[1 - p(H_k - H_{k-l})] \pmod{p^2}.$$

All denominators lie in $1, \dots, p-1$. Take fourth powers, set $r = k-\ell$, and sum to obtain (I.2.1).

Define additionally

$$R_p(m^{-1}) = \sum_{k<p} (4k+1) m^{-k} f_k, \quad E_p(m^{-1}) = \sum_{k<p} m^{-k} \sum_{r=0}^{k} (1+8r) \binom{k}{r}^4 (H_k - H_r).$$

The same substitution, with $n = p-1-r$ and $8n+7 = 8p-1-8r$, gives

$$\boxed{\sum_{n<p} (8n+7) S_n(m) \equiv m^{p-1}[-R_p(m^{-1}) + 8pB_p(m^{-1}) + 4pE_p(m^{-1})] \pmod{p^2}.} \tag{I.2.2}$$

Indeed, $\sum_r (1+8r)\binom{k}{r}^4 = (4k+1) f_k$, by $r \leftrightarrow k-r$. The harmonic correction has positive sign: the unperturbed weight and the first-order correction to the fourth power are both negative. The product of $8p$ with a first-order correction vanishes modulo p^2.

If $q_p(m) = (m^{p-1}-1)/p$, equation (I.2.1) gives the equivalence

$$\sum_{n<p} S_n(m) \equiv B_p(m^{-1}) \pmod{p^2} \quad \Longleftrightarrow \quad 4D_p(m^{-1}) \equiv q_p(m) B_p(m^{-1}) \pmod{p}. \tag{I.2.3}$$

Thus $D_p = T_p/4$ and $R_p = (1+4\theta)B_p$, with precisely the truncations fixed above. The finite-product proof is valid for p-adic units because its denominators are units.

□

I.3 The formal period and its logarithmic companion

Lemma I.3.1 (PFLOG). *Put $f_n = \sum_{j=0}^{n} \binom{n}{j}^4$, $d_n = \sum_{j=0}^{n} \binom{n}{j}^4 (H_n - H_j)$, $G(x) = \sum_{n\ge 0} f_n x^n$, and $D(x) = \sum_{n\ge 0} d_n x^n$. For $\theta = x\, d/dx$, set*

$$\mathcal{L} = \theta^3 - 2x(2\theta+1)(3\theta^2+3\theta+1) - 4x^2(\theta+1)(16(\theta+1)^2 - 1).$$

Then, as formal series with a formal log x,

$$\mathcal{L}G = 0, \qquad \mathcal{L}\big(G\log x + 4D\big) = 0.$$

Moreover, the latter solution is normalized by $G(0) = 1, D(0) = 0$. The resulting canonical formal coordinate is

$$q(x) = \exp((G \log x + 4D)/G) = x \exp(4D/G).$$

Proof.

I.3.1 All-n certificate (finite sums; no convergence assumptions)

For a complex or formal parameter s, use the polynomial generalized binomial $T(s,k) = \binom{s}{k}^4$ for integer $k \geq 0$. Define

$$A(s) = 2(2s+1)(3s^2+3s+1), \quad B(s) = 4s(16s^2-1),$$

and

$$\begin{aligned} r_s(k) = {} & (16s+4)k^5 - (104s^2+90s+16)k^4 \\ & + (276s^3+416s^2+184s+24)k^3 \\ & - (374s^4+818s^3+600s^2+172s+16)k^2 \\ & + (260s^5+756s^4+796s^3+368s^2+72s+4)k \\ & - 75s^6 - 275s^5 - 385s^4 - 255s^3 - 80s^2 - 10s. \end{aligned}$$

Set

$$C(s,k) = \frac{k^4 r_s(k)}{s^3(s+1)^4} \binom{s+1}{k}^4.$$

The exact rational identity is

$$(s+1)^3 T(s+1,k) - A(s)T(s,k) - B(s)T(s-1,k) = C(s,k+1) - C(s,k). \tag{I.3.1}$$

It is valid for s away from 0,-1, and extends wherever the sides are defined. To verify, divide by T(s,k) as a rational identity, use $T(s,k+1)/T(s,k) = (s-k)^4/(k+1)^4$, and clear denominators. The resulting polynomial identity is

$$(s+1-k)^4 r_s(k+1) - k^4 r_s(k) = s^3(s+1)^7 - s^3 A(s)(s+1-k)^4 - 4(16s^2-1)(s-k)^4(s+1-k)^4.$$

This explicit equality can be checked by expansion. `source_logperiod_check.py` reconstructs r with exact rational algebra and checks the residual symbolically as zero; output `source_logperiod_check.txt`, Python 3.13.6/SymPy 1.14.0. No finite agreement is being extrapolated.

Fix $n \geq 1$ and sum (I.3.1) over $0 \leq k \leq n+1$, keeping s variable. The certificate term $C(s,0)$ is zero. The term $C(s,n+2)$ is divisible by $(s-n)^4$ near $s = n$: its binomial factor contains $s-n$, and its denominator is nonzero there. Thus both the summed identity and its s-derivative vanish at $s = n$.

For every integer $m \geq 0$ and finite k,

$$\partial_s T(s,k)|_{s=m} = \begin{cases} 4\binom{m}{k}^4 (H_m - H_{m-k}), & k <= m, \\ 0, & k > m. \end{cases}$$

The second case follows from a fourth-order zero. Summing and replacing k by $m-k$ gives $4d_m$. Thus (I.3.1) gives both the ordinary recurrence

$$(n+1)^3 f_{n+1} = A(n) f_n + B(n) f_{n-1}$$

and its differentiated counterpart, with $h_m = 4d_m$,

$$(n+1)^3 h_{n+1} - A(n) h_n - B(n) h_{n-1} = -3(n+1)^2 f_{n+1} + A'(n) f_n + B'(n) f_{n-1}. \tag{I.3.2}$$

This covers every $n \geq 1$. At $n = 0$, the initial relations are $f_1 = 2f_0$ and $h_1 = 4$, with $f_0 = 1, f_1 = 2, h_0 = 0$, directly from the definitions.

For an operator polynomial $P(\theta)$, $P(\theta)(G \log x) = P(\theta) G \log x + P'(\theta) G$. Thus (I.3.2), together with those initial values, is exactly the coefficient equation for $\mathcal{L}(G \log x + 4D) = 0$. This proves the result for all coefficients.

□

I.4 Exact modular and inverse-coordinate normalization

Lemma I.4.1 (TATE)**.** *Let $Q = e^{2\pi i \tau}$ denote the Tate parameter, let $\eta_d = \eta(d\tau)$, and put*

$$X(\tau) = \frac{\eta_2^6 \eta_5^6}{\eta_1^6 \eta_{10}^6}, \qquad x(\tau) = \frac{X}{(X-1)^2},$$
$$P_d = E_2(Q^d), \qquad \Theta = \frac{10P_{10} + 5P_5 - 2P_2 - P_1}{12}.$$

The established identity of Shaun Cooper is

$$G(x(\tau)) = \Theta(\tau), \qquad G(x) = \sum_{n \geq 0} \left(\sum_j \binom{n}{j}^4 \right) x^n. \tag{I.4.1}$$

Reference: S. Cooper, Sporadic sequences, modular forms and new series for 1/pi*, Ramanujan J. 29 (2012), 163–183, equations (9), (23)–(24); author PDF. The Eisenstein expression displayed here fixes the normalization used throughout this proof.*

Define $\psi = L/G$ and use the series ℓ_n, L, q fixed above. Then

$$\boxed{q(x(Q)) = Q.} \tag{I.4.2}$$

This is an exact formal identity over $\mathbb{Q}$, equivalently an analytic identity near the cusp. In particular the canonical coordinate used in Lemma I.5.1 is the actual inverse of Cooper's eta-defined parameter, with leading coefficient one, not an unspecified rescaling or power of it.

Proof.

I.4.1 An exact modular differential identity

Write $\delta = Q\, d/dQ$ and

$$A = \delta \log X = \frac{2P_2 + 5P_5 - P_1 - 10P_{10}}{4}.$$

The eta product derivative gives this equality exactly. We prove

$$F := \Theta^2 (X^2 - 18X + 1) - (X-1)^2 A^2 = 0. \tag{I.4.3}$$

Both Θ and A are holomorphic modular forms of weight two and trivial character on $\Gamma_0(10)$. One standard verification is to use $E_{2,d} = E_2(Q) - dE_2(Q^d)$, each a holomorphic weight-two modular form on $\Gamma_0(d)$, and

$$\Theta = (E_{2,2} - E_{2,5} - E_{2,10})/12, \qquad A = -(E_{2,2} + E_{2,5} - E_{2,10})/4.$$

An alternative verification avoids needing that Eisenstein assertion: put

$$Y = \frac{\eta_5^2 \eta_{10}^2}{\eta_1^2 \eta_2^2}.$$

Its weight is zero, the two eta-criterion sums are $24, -24$, and $\prod d^{r_d} = 625$ is a rational square. Thus Y, like X, is a modular unit with trivial character, and $\Theta = \delta \log Y$ exactly. The logarithmic derivative of a weight-zero modular unit is a weight-two modular form; it is holomorphic in the upper half-plane because the unit has no zeros or poles there. At every cusp its logarithm has the form $r \log Q_c +$ a convergent regular series; its weight-two logarithmic derivative is consequently holomorphic at that cusp as well. This proves the needed modularity and holomorphicity of both $A = \delta \log X$ and $\Theta = \delta \log Y$ directly from eta-quotient modularity.

The eta-quotient criterion gives X as a weight-zero modular function with trivial character on $\Gamma_0(10)$. For the exponent vector $(r_1, r_2, r_5, r_{10}) = (-6, 6, 6, -6)$, the two criterion sums are

$$\sum d r_d = -24, \qquad \sum (10/d) r_d = -24,$$

both multiples of 24; and $\prod d^{r_d} = 1$, giving trivial character. Eta has no zeros in the upper half-plane, so possible poles of X occur only at cusps.

For completeness, the cusp-order formula used here, for a denominator $c \mid 10$, is

$$\operatorname{ord}_{1/c}\left(\prod_{d|10} \eta_d^{r_d}\right) = \frac{10}{24 \gcd(c^2, 10)} \sum_{d|10} \frac{r_d \gcd(c, d)^2}{d}.$$

Orders are measured in the local parameter at that cusp. Since 10 is squarefree, these four denominator classes are all the cusps. The formula gives

cusp denominator c	1	2	5	10 (infinity)
$\operatorname{ord} X$	-1	1	1	-1
$\operatorname{ord} E$, $E = \eta_1^4 \eta_2^4 \eta_5^4 \eta_{10}^4$	3	3	3	3

For E, the exponent sum is 16, so the weight is eight; both criterion sums are 72, and the character is trivial since $\prod d^4 = 100^4$ is a square. It is holomorphic at all cusps and in the upper half-plane. Therefore EF is a holomorphic modular form of weight twelve and trivial character on $\Gamma_0(10)$: the only possible poles of F have order at most two at denominator classes 1 and 10, and E's order three cancels them.

The index is

$$[\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(10)] = 10(1 + 1/2)(1 + 1/5) = 18.$$

The standard modular-form valence bound therefore says a nonzero holomorphic weight-twelve form on this group has order at infinity at most $12 \cdot 18/12 = 18$. Consequently the **finite exact check** of its coefficients of $Q^0, \ldots, Q^{18}$ establishes (I.4.3) if all vanish.

That complete check is supplied by `algebra_tate_normalization_check.py`, using exact integer and rational arithmetic, with its output `algebra_tate_normalization_check.txt` and coefficient certificate `algebra_tate_normalization_certificate.json`. The script

expands the specified eta products and Eisenstein series, not fitted or inferred series. To remove negative powers, it writes $X = Q^{-1}U_X$, $E = Q^3U_E$, and calculates the identity

$$EF = Q\left[U_E\Theta^2(U_X^2 - 18QU_X + Q^2) - U_EA^2(U_X - Q)^2\right].$$

Every coefficient through Q^{18}, inclusive, is exactly zero. The JSON lists all coefficients through the required bound for U_X, U_E, Θ, A, and EF, so the finite certificate can be independently checked. The valence argument is the rigorous reduction from the general identity to that finite check.

The eta-quotient modularity criterion, its cusp formula, and the valence bound are standard established modular-form results. Exact source statements were checked in the following primary author sources: J. Rouse and J. J. Webb, *On spaces of modular forms spanned by eta-quotients*, arXiv:1311.1460v3, section 1, printed page 2 (Newman's criterion and Ligozat's cusp-order theorem); S. Zerbes, *Modular Forms*, ETH lecture notes, spring 2021/22, Theorem 2.6.3 and Corollary 2.6.8, printed pages 35–38 (valence and its finite coefficient consequence). The former's cusp formula denominator $\gcd(c, 10/c)c$ equals $\gcd(c^2, 10)$, giving precisely the displayed formula. Their exact hypotheses and all relevant inputs are displayed above; the proof does not rely on any unstated experimental modular identification.

Now

$$\delta \log x = -\frac{X+1}{X-1}A,$$

and direct rational algebra gives

$$(1 - 16x)(1 + 4x) = \frac{(X^2 - 18X + 1)(X + 1)^2}{(X - 1)^4}.$$

Thus (I.4.3) proves

$$(\delta \log x)^2 = \Theta^2(1 - 16x)(1 + 4x).$$

Since $X = Q^{-1} + 6 + O(Q)$, $x = Q - 4Q^2 + O(Q^3)$, and $\Theta = 1 + O(Q)$, the sign of the square root at the cusp is fixed:

$$\delta \log x = \Theta\sqrt{(1 - 16x)(1 + 4x)}, \tag{I.4.4}$$

where the square root has constant term one.

I.4.2 The period's logarithmic coordinate has the same derivative

The complete logarithmic-period proof Lemma I.3.1 establishes

$$\mathcal{L}G = 0, \qquad \mathcal{L}(G \log x + L) = 0,$$

with

$$\mathcal{L} = \theta^3 - 2x(2\theta + 1)(3\theta^2 + 3\theta + 1) - 4x^2(\theta + 1)(16(\theta + 1)^2 - 1).$$

Write $\Delta = 1 - 12x - 64x^2 = (1 - 16x)(1 + 4x)$. Expanding gives

$$\mathcal{L} = \Delta\theta^3 + \tfrac{3}{2}(\theta\Delta)\theta^2 - (10x + 188x^2)\theta - (2x + 60x^2).$$

It is exactly the symmetric-square operator of

$$\theta^2 u + P\theta u + Ru = 0, \qquad P = \frac{\theta\Delta}{2\Delta}, \quad R = -\frac{x + 15x^2}{\Delta}. \tag{I.4.5}$$

This can be verified without any modular theorem: the symmetric-square coefficients are

$$3P, \quad \theta P + 4R + 2P^2, \quad 2\theta R + 4PR,$$

which give exactly the three displayed normalized coefficients of $\mathcal{L}$.

Let u be the unique formal analytic solution of (I.4.5) with constant term one, and let $v = u\log x +$ a formal analytic correction whose constant term is zero. Frobenius recurrences with leading factors n^2 give existence and uniqueness in characteristic zero. Then $u^2 = G$, by the unique normalized analytic solution of the third-order equation. Similarly $uv = G\log x + L$: both have the same logarithmic coefficient G, solve the same third-order equation, and have zero constant term in their analytic correction. Their difference is an analytic multiple of the unique normalized solution G, and its constant term fixes that multiple to zero.

The Wronskian $W = u\theta v - v\theta u$ satisfies $\theta W = -PW$. Its normalization at zero is one, so

$$W = \Delta^{-1/2}.$$

Since $\log q = \log x + L/G = v/u$, it follows that

$$\theta \log q = \frac{W}{u^2} = \frac{1}{G(x)\sqrt{\Delta(x)}}. \tag{I.4.6}$$

I.4.3 Matching the normalized coordinates

Cooper's (I.4.1) and (I.4.4) imply that the formal inverse $Q = Q(x)$, normalized by $Q = x + O(x^2)$, satisfies

$$\theta_x \log Q = \frac{1}{G(x)\sqrt{\Delta(x)}}.$$

By (I.4.6), the same equation holds for $q(x)$. Their quotient is constant in characteristic zero. Both have leading term x, so that constant is one. Hence $Q(x) = q(x)$, proving (I.4.2).

I.4.4 Exact eta-parameter bridge

For this bridge write $r = X$ and $z = \eta_2\eta_{10}^3/(\eta_1^3\eta_5)$. We prove

$$rz = 1 + 9z + 20z^2 = (1 + 4z)(1 + 5z). \tag{I.4.7}$$

The eta-quotient criterion makes r, z weight-zero functions of trivial character on $\Gamma_0(10)$. Their cusp orders at denominators $1, 2, 5, 10$ are $(-1, 1, 1, -1)$ and $(-1, 0, 0, 1)$, respectively, by the cusp formula above. Hence $rz - 1 - 9z - 20z^2$ has a possible pole of order at most two only at denominator one. Multiplication by $E = \eta_1^4\eta_2^4\eta_5^4\eta_{10}^4$, of weight eight and cusp order three at every cusp, gives a holomorphic form of trivial character. Since $[\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(10)] = 18$, a nonzero such form has order at infinity at most $8 \cdot 18/12 = 12$.

The exact program `root_tate_bridge.py` expands these eta products and checks that every coefficient of $E(rz - 1 - 9z - 20z^2)$ through degree twelve in Q vanishes. Together with the valence bound this proves (I.4.7).

Put $N = z(1+4z)(1+5z)$ and $Q_z = 1+8z+20z^2$. Equation (I.4.7) gives $r-1 = Q_z/z$, hence $r/(r-1)^2 = N/Q_z^2$, the coordinate used in Lemma I.6.1. Thus the Tate lift $x(Q) \mapsto x(Q^p)$ is exactly $\sigma = q^{-1}(q(x)^p)$, where the inverse is compositional. The eta products give

$x(Q) \in Q + Q^2\mathbb{Z}[[Q]]$, so the inverse series and $\sigma = x(Q(x)^p)$ are integral over $\mathbb{Z}_p$ in all degrees. Moreover $\sigma/x^p \equiv 1 \pmod{p}$, with constant term one.

□

I.5 The first p-squared coefficients of the canonical correction

Lemma I.5.1 (DIGIT)**.** *For nonnegative integer n, put*

$$f_n = \sum_{j=0}^{n} \binom{n}{j}^4, \quad \ell_n = 4\sum_{j=0}^{n} \binom{n}{j}^4 (\mathcal{H}_n - \mathcal{H}_j), \quad \mathcal{H}_n = \sum_{r=1}^{n} \frac{1}{r}, \quad \mathcal{H}_0 = 0.$$

Set

$$G(x) = \sum_{n\geq 0} f_n x^n, \quad L(x) = \sum_{n\geq 0} \ell_n x^n, \quad \psi = L/G, \quad q = x\exp\psi.$$

For a prime $p \geq 5$, define

$$B = \sum_{b=0}^{p-1} f_b x^b, \qquad T = \sum_{b=0}^{p-1} \ell_b x^b, \qquad h = T/B, \qquad \sigma = q^{-1}(q(x)^p).$$

All operations are formal over $\mathbb{Q}_p$; $G(0) = 1$, $\psi(0) = 0$, and $q = x + O(x^2)$ make them well-defined. Then $z = \sigma/x^p$ is p-integral in all degrees strictly less than p^2, and

$$\boxed{z \equiv 1 + ph \pmod{p^2, x^{p^2}}.} \tag{I.5.1}$$

Thus the first p^2 coefficients (degrees $0, \ldots, p^2-1$) of the first-order canonical correction $(z-1)/p$ equal those of T/B modulo p. No global p-integrality of q is assumed or needed.

Proof.

I.5.1 Two direct block congruences

For all integers $0 \leq a, b < p$,

$$\begin{aligned} f_{ap+b} &\equiv f_a f_b + pa f_a \ell_b \pmod{p^2}, \\ p\ell_{ap+b} &\equiv f_b \ell_a + p f_a \ell_b + pa\ell_a \ell_b \pmod{p^2}. \end{aligned} \tag{I.5.2}$$

The second line is proved directly below. No differentiation of an integer-index congruence is being performed. The bounded assumption $a < p$ is indispensable to the precision argument, because it keeps $\mathcal{H}_a$ and ℓ_a p-integral.

For $0 \leq c \leq a$, $0 \leq d \leq b$, separate multiples of p from factorials. The unit factors in each full block of length p obey

$$\prod_{r=1}^{p-1} (up + r) \equiv (p-1)!\,(1 + up\mathcal{H}_{p-1}) \equiv (p-1)! \pmod{p^2},$$

because $\mathcal{H}_{p-1} \equiv 0 \pmod{p}$. The final partial blocks contribute the usual first-order harmonic factors. Hence

$$\binom{ap+b}{cp+d} \equiv \binom{a}{c}\binom{b}{d}\left[1 + p(a\mathcal{H}_b - c\mathcal{H}_d - (a-c)\mathcal{H}_{b-d})\right] \pmod{p^2}. \tag{I.5.3}$$

All the divisions used here are by p-adic units, since all digits are less than p. If $d > b$ and $cp + d \le ap + b$, subtraction in the lower base-p digit borrows once, so the binomial coefficient is divisible by p. This also follows directly from the number of multiples of p in its factorials, since $a < p$. Such a term has fourth power divisible by p^4.

The harmonic sums have the exact precision

$$p\mathcal{H}_{ap+b} \equiv \mathcal{H}_a + p\mathcal{H}_b \pmod{p^2}. \tag{I.5.4}$$

Indeed the multiples of p contribute $\mathcal{H}_a$ after multiplication by p. Each full block of unit denominators contributes zero modulo p^2 after that multiplication, using $\mathcal{H}_{p-1} \equiv 0 \pmod{p}$; the partial block contributes $p\mathcal{H}_b$. Thus

$$p(\mathcal{H}_{ap+b} - \mathcal{H}_{cp+d}) \equiv \mathcal{H}_a - \mathcal{H}_c + p(\mathcal{H}_b - \mathcal{H}_d) \pmod{p^2}. \tag{I.5.5}$$

Every expression on the right is p-integral. In the discarded borrow terms the factor on the left is also p-integral, so the fourth-power factor still makes their contribution zero modulo p^2.

Put

$$F_{ac} = \binom{a}{c}^4, \quad F_{bd} = \binom{b}{d}^4, \quad R_{cd} = a\mathcal{H}_b - c\mathcal{H}_d - (a-c)\mathcal{H}_{b-d}.$$

Binomial symmetry in d gives

$$\sum_{d=0}^{b} F_{bd}R_{cd} = a\sum_{d=0}^{b} F_{bd}(\mathcal{H}_b - \mathcal{H}_d) = \frac{a}{4}\ell_b, \tag{I.5.6}$$

independently of c. Summing the fourth power of (I.5.3) and using (I.5.6) proves the first line of (I.5.2).

For the second, multiply that fourth power by (I.5.5). Its contribution to $p\ell_{ap+b}/4$, summed over c, d, is

$$\sum_{c,d} F_{ac}F_{bd}\left[(\mathcal{H}_a - \mathcal{H}_c) + p(\mathcal{H}_b - \mathcal{H}_d) + 4pR_{cd}(\mathcal{H}_a - \mathcal{H}_c)\right] \pmod{p^2}.$$

The three sums respectively equal

$$\frac{1}{4}f_b\ell_a, \qquad \frac{p}{4}f_a\ell_b, \qquad \frac{pa}{4}\ell_a\ell_b,$$

the last by (I.5.6). Multiplication by four proves the second line of (I.5.2).

I.5.2 A finite logarithmic Frobenius relation

From now on work modulo x^{p^2}. Write

$$X = G(x^p), \qquad Y = L(x^p), \quad \dot{X} = (\theta G)(x^p), \quad \dot{Y} = (\theta L)(x^p), \qquad \theta = x\frac{d}{dx}.$$

The dots mean Euler differentiation **before** replacing x by x^p; there is no unwanted extra factor of p. Only coefficients with upper digit $a < p$ occur. In particular $X, Y, \dot{X}, \dot{Y}$ are p-integral to this precision, and $X(0) = 1$. Summing (I.5.2) over the digit decomposition $n = ap + b$ gives

$$\begin{aligned} G(x) &\equiv BX + pT\dot{X} \pmod{p^2, x^{p^2}}, \\ pL(x) &\equiv BY + pT(X + \dot{Y}) \pmod{p^2, x^{p^2}}. \end{aligned} \tag{I.5.7}$$

The coefficients of L at indices below p^2 have valuation at least -1, because the only denominators in $\mathcal{H}_n$ divisible by p have exactly one factor of p in this range. Thus pL is p-integral; all quotient manipulations in the next calculation are modulo p^2 between p-integral series.

Divide the second congruence in (I.5.7) by the first. Since $h = T/B$,

$$\begin{aligned} p\psi(x) &\equiv \frac{Y + ph(X + \dot{Y})}{X + ph\dot{X}} \\ &\equiv \frac{Y}{X} + ph\left(1 + \frac{\dot{Y}}{X} - \frac{Y\dot{X}}{X^2}\right) \\ &= \psi(x^p) + ph(1 + (\theta\psi)(x^p)) \pmod{p^2, x^{p^2}}. \end{aligned} \tag{I.5.8}$$

Consequently

$$\Lambda = p\psi(x) - \psi(x^p) \in p\mathbb{Z}_p[[x]]/(x^{p^2}), \qquad \Lambda/p \equiv h(1 + (\theta\psi)(x^p)) \pmod{p, x^{p^2}}. \tag{I.5.9}$$

I.5.3 Canonical inverse and bounded integrality

For $0 \le n < p$, the coefficients of ψ are p-integral. Therefore the coefficients of $\exp\psi$ through degree $p-1$ are p-integral: the factorials in the exponential have index at most $p-1$. Equivalently,

$$q_n \in \mathbb{Z}_p \quad (1 \le n \le p).$$

Hence $\exp\psi(x^p)$, $(\theta\psi)(x^p)$, and $q'(x^p)$ are p-integral modulo x^{p^2}, and their relevant constant terms are one except for the Euler derivative, whose constant term is zero.

Define $z = \sigma/x^p$ formally over $\mathbb{Q}_p$. The identity $q(\sigma) = q(x)^p$, after division by x^p, says

$$\sum_{n=1}^{p} q_n x^{p(n-1)} z^n = \exp\psi(x^p)\,\exp\Lambda \pmod{x^{p^2}}. \tag{I.5.10}$$

No coefficients q_n with $n > p$ occur because their minimal degree after division is $p(n-1) \ge p^2$. The left side has leading term z; its other terms have a factor x^p. The right side is p-integral, by (I.5.9) and the preceding paragraph. Recursive coefficient solution of (I.5.10) therefore makes z p-integral through degree $p^2 - 1$.

Moreover $\exp\Lambda \equiv 1 \pmod{p}$. Thus reducing (I.5.10) modulo p, its right side is the value of its left side at $z = 1$. The difference between the left side at z and at one factors as $(z-1)$ times a unit with constant term one. Therefore $z \equiv 1 \pmod{p}$. Write $z = 1 + pv$, with v p-integral to this range.

For every $m \ge 2$, $v_p(p^m/m!) \ge 2$, so $\exp\Lambda \equiv 1 + \Lambda \pmod{p^2}$. Taylor expansion of the polynomial left side of (I.5.10), whose coefficients are p-integral, is consequently legitimate modulo p^2, and gives

$$pv\,q'(x^p) \equiv \exp\psi(x^p)\,\Lambda \pmod{p^2, x^{p^2}}.$$

Use the exact identity

$$q'(x^p) = \exp\psi(x^p)(1 + (\theta\psi)(x^p)).$$

Both factors on the right are units in the truncated p-integral power-series ring. Divide by p and by these units, then use (I.5.9), to obtain $v \equiv h \pmod{p, x^{p^2}}$. This is (I.5.1).

□

I.6 Rational canonical correction and its pole bound

Lemma I.6.1 (POLES)**.** *Fix a prime $p \geq 7$. Define the cover coordinate and two polynomials*

$$z = \frac{\eta_2 \eta_{10}^3}{\eta_1^3 \eta_5}, \quad N(z) = z(1+4z)(1+5z), \quad Q_z(z) = 1 + 8z + 20z^2.$$

Lemma I.4.1 gives $x = N/Q_z^2 = X/(X-1)^2$. The integral Tate lift $\sigma(x(Q)) = x(Q^p)$ has an expansion $\sigma = x^p(1 + p h_{\mathrm{can}}) \pmod{p^2}$. Then $h_{\mathrm{can}} \in \mathbb{F}_p(x) \cap x\mathbb{F}_p[[x]]$, each of its finite poles is simple and is a zero of B_p, and its pole order at infinity is at most $p+1$. Consequently $B_p h_{\mathrm{can}}$ is a polynomial of degree less than $2p$.

Proof. Here is the polynomial degree estimate needed for the last assertion. For $n < p$ put $r_0 = p - 1 - n$. For every $j \in \mathbb{F}_p$, extending the left side by zero for $j > n$,

$$\binom{n}{j} \equiv (-1)^j \binom{p-1-j}{r_0} \pmod{p}.$$

After taking fourth powers the right side is a polynomial of degree $4r_0$ in j. If $4r_0 < p-1$, its sum over $\mathbb{F}_p$ vanishes, since each monomial degree is less than $p-1$, including the constant term whose sum is $p = 0$. Therefore

$$\deg B_p \leq \left\lfloor \frac{3(p-1)}{4} \right\rfloor. \tag{I.6.1}$$

The modular and integral-geometric proof of the pole assertion follows. In its formulas the polynomial denoted $Q(z)$ is $Q_z(z)$ above, and the formal first correction denoted h is h_{can}.

I.6.1 Geometric inputs

- P. Deligne and M. Rapoport, *Les schémas de modules de courbes elliptiques*, LNM 349 (1973), pp. 143–316, author-hosted complete PDF. Use VI Proposition 6.7, printed pp. 285–286, for smooth coarse models away from the level; VI Theorem 6.9 and its proof, pp. 286–287, for the two characteristic-p components and their degeneracy maps; VII Theorems 3.9–3.10 and Corollary 3.13, pp. 304–306, for integral Fourier expansions. VI §6.8 and the proof of 6.9 explicitly retain arbitrary prime-to-p level K.
- N. Katz, *p-adic properties of modular schemes and modular forms*, LNM 350 (1973), pp. 69–190, author-hosted complete PDF. Use §2.0 for the Hasse invariant and its Q-expansion; Theorem 3.1 and §3.4, pp. 112–114, for the canonical subgroup, including its Tate-curve value; §3.10, pp. 122–123, for the resulting Frobenius morphism. The discussion immediately preceding Theorem 4.3 bis, p. 149, states the simple-zero theorem for the Hasse invariant on a fine modular curve.

These references establish the stated basic geometry. The particular quotient, divisor, and pole arguments below are deductions, not claims that the references print this lemma. No supercongruence precision theorem from Beukers 2403.03301 is used.

I.6.2 The coordinate x has a good integral model

Work over R = $\mathbb{Z}_p$. The standard eta transformation criterion gives that z has weight zero, trivial character, and level 10: the exponent vector for divisors 1,2,5,10 is (-3,1,-1,3), the two eta congruence sums are 24 and -24, and the character parameter is 400, a square. The cusp orders, in the order with cusp denominators 1,2,5,10, are

$$(-1, 0, 0, 1). \tag{I.6.2}$$

These follow directly from

$$\operatorname{ord}_{a/c} z = \frac{10}{24\gcd(c^2,10)} \sum_{d|10} r_d \frac{\gcd(c,d)^2}{d}.$$

Thus z has one simple horizontal pole and one simple horizontal zero. There is no vertical divisor: on the interior, $z^{24} = \Delta_2\Delta_{10}^3/(\Delta_1^3\Delta_5)$ is a modular unit, up to the harmless differential-normalization factors, which are powers of 2 and 5. At a vertical generic point its valuation is zero, also seen from the unit leading coefficient of its Fourier expansion at the cusp. The special fiber of $X_0(10)_R$ is irreducible and smooth. The two relevant cusp sections are disjoint. Therefore z gives a degree-one map on each fiber, and

$$X_0(10)_R \simeq \mathbb{P}^1_{R,z}. \tag{I.6.3}$$

For completeness, the degree-four quotient map itself can be checked without integral-model guesses. Its three nonidentity deck transformations are

$$z \mapsto \frac{1}{20z}, \qquad z \mapsto -\frac{4z+1}{20z+4}, \qquad z \mapsto -\frac{5z+1}{20z+5}. \tag{I.6.4}$$

They have invertible determinants over R, are distinct involutions, form a group of order four, and fix x. The resultant

$$\operatorname{Res}_z(N, Q^2) = 400 \tag{I.6.5}$$

is a unit. The homogenized numerator and denominator thus give a degree-four morphism $\mathbb{P}^1_R \to \mathbb{P}^1_R$ on every fiber. Here is an explicit eta-law check that the invariant map is the full Atkin–Lehner quotient. Take

$$W_2 = \begin{pmatrix} 2 & -1 \\ 10 & -4 \end{pmatrix}.$$

For $d = 1, 2, 5, 10$, respectively, express $dW_2\tau$ as the following determinant-one matrix applied to $d'\tau$:

d	d'	matrix	eta multiplier exponent in $e^{\pi i r}$
1	2	$\begin{pmatrix} 1 & -1 \\ 5 & -4 \end{pmatrix}$	$-1/2$
2	1	$\begin{pmatrix} 2 & -1 \\ 5 & -2 \end{pmatrix}$	$-1/4$
5	10	$\begin{pmatrix} 1 & -5 \\ 1 & -4 \end{pmatrix}$	$-1/2$
10	5	$\begin{pmatrix} 2 & -5 \\ 1 & -2 \end{pmatrix}$	$-1/4$

These exponents follow directly from the usual eta law $r = (a+d)/(12c) - s(d,c) - 1/4$ for $c > 0$, where s is the Dedekind sum; here $s(-4,5) = 1/5$, $s(-2,5) = 0$, and sums with denominator 1 are zero. The square-root factors are $10\tau - 4, 5\tau - 2, 10\tau - 4, 5\tau - 2$ to the power $1/2$. In the sixth-power quotient defining X, both these factors and all multipliers cancel. Consequently $X(W_2\tau) = 1/X(\tau)$. Eta inversion likewise gives $X(-1/(10\tau)) = X(\tau)$. Hence $x = X/(X-1)^2$ is invariant under both w_2 and w_{10}, which generate the order-four Atkin–Lehner group. Since the map already has degree four, its quotient is precisely that group, whose three nonidentity deck maps are (I.6.4). Thus the displayed invariant map is the coarse Atkin–Lehner quotient. Hence

$$X_R^* \simeq \mathbb{P}^1_{R,x}. \tag{I.6.6}$$

There is no loss of degree on reduction for p $\geq$ 7. The map is tame. Explicitly,

$$\begin{gathered}\frac{dx}{dz} = -\frac{(20z^2-1)(20z^2+10z+1)}{Q^3}, \\ 16N - Q^2 = -(20z^2-1)^2, \quad 4N + Q^2 = (20z^2+10z+1)^2.\end{gathered} \tag{I.6.7}$$

The three branch values of this fourfold quotient map are $1/16, -1/4, \infty$. Their branch quadratics have discriminants 80,20,-16. None collides or becomes inseparable for p $\geq$ 7. This accounts explicitly for the extra quotient ramification. One must still use a fine cover for modular-form valuations; treating these coarse coordinates as fine moduli parameters would be incorrect.

I.6.3 An integral Hecke polynomial and its exact reduction

Let C be the compact modular curve $X_0(10p)_R/\langle w_2, w_5\rangle$, with its two degeneracy maps to X_R^*. The Atkin–Lehner operations use only degrees 2 and 5, so they act on the integral model and commute with the p-isogeny square. Each projection has degree p + 1 generically.

The pair of maps is generically one-to-one onto its image. For a generic elliptic curve E, $\mathrm{End}(E) = \mathbb{Z}$. If two cyclic p-quotients $E/D_1, E/D_2$ are identified by an Atkin–Lehner operation of degree $d \in \{1, 2, 5, 10\}$, composing the two p-isogenies, the degree-d identification, and a dual isogeny gives an endomorphism of degree $p^2 d$. For $d = 2, 5, 10$, this is not a square, a contradiction. For $d = 1$, the composite is $\pm[p]$; dual-isogeny cancellation forces the same original kernel. Thus $D_1 = D_2$, and the $p + 1$ generic images are distinct.

Let D be the horizontal image divisor in $S = \mathbb{P}^1_{R,X} \times \mathbb{P}^1_{R,Y}$. Since S is regular, the horizontal prime divisor D is Cartier. Its two generic projection degrees show that its divisor class is $\mathcal{O}_S(p+1, p+1)$. Thus it has a primitive bihomogeneous defining equation, whose affine version satisfies

$$\Phi_p(X, Y) \in R[X, Y], \quad \deg_X \Phi_p, \deg_Y \Phi_p \leq p+1, \quad \Phi_p(x(Q), x(Q^p)) = 0. \tag{I.6.8}$$

Here is the reduction argument including the coarse quotient. By Deligne–Rapoport VI 6.9, the special fiber before the order-four quotient has two reduced components with degeneracy maps (id,Frob) and (Frob,id). The prime-to-p Atkin–Lehner operations preserve the two subgroup types. On a $\Gamma 0(10)$ moduli problem the possible diamond-p operation is trivial. On each component their quotient is $X^*_{\mathbb{F}_p}$. Taking invariants under a group of order four commutes with reduction, by the averaging projector $\frac{1}{4}\sum_g g$. Therefore the special fiber of C has precisely those two components after quotient, and its proper image has support exactly the two graphs

$$Y = X^p, \qquad X = Y^p. \tag{I.6.9}$$

One does not need to assume that a singular image is normal, or assert that its normalization commutes with reduction. The map C $\to$ D is proper and surjective; it stays surjective on the special fiber. Thus (I.6.9) gives the support of the special-fiber divisor of the primitive equation. The two graphs have bidegrees (p,1) and (1,p). Their sum already has bidegree (p+1,p+1). There is consequently neither an extra component nor a repeated graph. Rescaling Φ_p by an R-unit gives

$$\Phi_p(X,Y) = (X^p - Y)(X - Y^p) + p\Psi_p(X,Y), \quad \Psi_p \in R[X,Y], \quad \deg_X \Psi_p, \deg_Y \Psi_p \le p+1. \tag{I.6.10}$$

This is the required Kronecker congruence for this coordinate, proved from the integral correspondence. It is not imported merely from the classical j-polynomial.

I.6.4 Rationality, simple finite poles, and the infinity bound

The eta expression gives $x(Q) \in Q + Q^2\mathbb{Z}[[Q]]$. Its compositional inverse is integral too. Hence the formal Tate lift lies in $x^p\mathbb{Z}_p[[x]]$, is congruent to x^p modulo p, and has coefficient one at degree p. Consequently $h \in x\mathbb{F}_p[[x]]$ is well defined.

Write $\sigma = x^p + p\delta$. Substitution into (I.6.10), modulo p^2, gives

$$0 = -p\delta(x - x^{p^2}) + p\Psi_p(x, x^p),$$

so in the field $\mathbb{F}_p((x))$,

$$\delta = \frac{\overline{\Psi}_p(x,x^p)}{x - x^{p^2}}, \qquad h = \frac{\overline{\Psi}_p(x,x^p)}{x^p(x - x^{p^2})}. \tag{I.6.11}$$

This proves rationality. The polynomial $x - x^{p^2}$ is squarefree, since its derivative is one. Thus h has at most a simple pole at each nonzero finite point. At $x = 0$, the formal-series construction cancels the entire apparent denominator, including x^p.

The numerator in (I.6.11) has degree at most $(p+1) + p(p+1) = (p+1)^2$, while its displayed denominator has degree $p^2 + p$. Therefore

$$\operatorname{poleord}_\infty(h) \le (p+1)^2 - (p^2+p) = p+1. \tag{I.6.12}$$

Cancellation can only improve this inequality. The coefficient of $X^{p+1}Y^{p+1}$ is allowed to vanish or not vanish. No leading-coefficient assumption is hidden in (I.6.12).

I.6.5 Why every ordinary finite pole cancels

On the formal ordinary locus, quotienting by the canonical subgroup gives a Frobenius lift. On Tate(Q) its value is Tate(Q^p), so its formal branch is exactly σ.

This construction descends through the Atkin–Lehner quotient. Indeed, if $\alpha : E \to E'$ is a prime-to-p isogeny between ordinary families, the image of the canonical subgroup is the canonical subgroup of E': this follows from the uniqueness of the subgroup lifting the characteristic-p Frobenius kernel. Thus the two quotient operations commute. The same averaging projector as above gives descent of regular functions through the tame order-four quotient and commutes with p-adic completion.

At a finite ordinary point, x is regular at the point and its Frobenius image. The canonical lift $\sigma(x)$ is regular and reduces to x^p, so $(\sigma(x) - x^p)/p$ is regular modulo p in the formal ordinary ring. It equals the rational function δ of (I.6.11), since their expansions agree at the ordinary cusp. Hence δ has no ordinary finite pole, and neither does h away from the already treated point $x = 0$.

This proves the actual cancellation. It is not an inference just from the factor $x - x^{p^2}$, whose other zeros can certainly be ordinary points.

I.6.6 The Hasse relation covers the supersingular poles, including elliptic points

Use a connected fine cover with auxiliary full level 3, retaining Γ0(10) level, after an unramified extension if needed. This is allowed because p ≥ 7. On this smooth curve let A be the Hasse invariant, a section of ω^{p-1}, normalized by its Q-expansion 1. It has a simple zero at every supersingular point.

The form Θ of Lemma I.4.1 is a holomorphic weight-two modular form for Γ0(10). Its Fourier coefficients are integral at p ≥ 7, and the integral Q-expansion principle makes it an integral section on this cover. Equivalently, writing $E_{2,d} = E_2(Q) - dE_2(Q^d)$, it is $(E_{2,2} - E_{2,5} - E_{2,10})/12$; each difference is holomorphic of weight two. The denominator 12 causes no problem for the stated primes.

Elementary Lucas gives $f_{pa+b} \equiv f_a f_b \pmod{p}$ for $0 \le b < p$. Therefore

$$G(x) = B(x)G(x^p) \quad \text{in } \mathbb{F}_p[[x]].$$

After substituting x(Q) and using (I.4.1), this becomes $\Theta = B(x)\Theta^p$. The ratio of the same-weight sections A^2 and Θ^{p-1} has Q-expansion Θ^{1-p}. Injectivity of expansion in the completed function field gives the exact rational-function identity on the fine curve

$$B(x) = \frac{A^2}{\Theta^{p-1}}. \tag{I.6.13}$$

No assumption about an Atkin–Lehner character of A is needed: this equality is first proved upstairs, where all sections are honest sections of the stated powers of ω. Its left side already descends to the x-line.

Now let P lie above a finite supersingular point x = a. Choose a nonvanishing local generator of ω and write $m = \operatorname{ord}_P \Theta$, an integer ≥ 0. Since B(x) is a polynomial, it has no pole at P. Formula (I.6.13) gives

$$0 \le \operatorname{ord}_P B(x) = 2 - (p-1)m. \tag{I.6.14}$$

For p ≥ 7 this forces m = 0 and $\operatorname{ord}_P B(x) = 2$. Thus every finite supersingular point is a zero of B. If e is the ramification index of the map to the coarse x-line, then

$$e \operatorname{ord}_{x=a} B = 2. \tag{I.6.15}$$

This proves directly that the coarse zero has multiplicity 1 or 2 and handles elliptic ramification. At a finite ordinary point the same calculation has 0 in place of 2; regularity forces m = 0 and B is a unit. Consequently the finite zero set of B is exactly the supersingular set.

The argument deliberately makes no assertion that a Hasse section itself is a coarse function. Taking its simple zero order on the coarse line without the factor e would be an error. Notice also that (I.6.14) proves that Θ has no zero over any finite x even in the special fiber, without assuming that its complex zero divisor specializes without collision.

Combining §§5–7, every finite pole of h is simple and is killed by B. Therefore Bh has no finite poles on the projective x-line, hence is a polynomial. The infinity bound (I.6.12) gives the stated polynomial degree bound, completing the intermediate proof.

□

I.7 Global first canonical correction

Lemma I.7.1 (CAN). *For every $p \ge 7$, with $\sigma = q^{-1}(q(x)^p)$, one has*

$$\frac{\sigma(x)}{x^p} \equiv 1 + p\frac{T(x)}{B(x)} \pmod{p^2} \quad \textit{in } \mathbb{Z}_p[[x]], \tag{I.7.1}$$

and, as rational functions in $\mathbb{F}_p(x)$,

$$\theta(T/B) = \frac{\Delta^{(p-1)/2}}{B} - 1. \tag{I.7.2}$$

Proof.

I.7.1 The coefficient and degree argument

First, $\deg B_p \le \lfloor 3(p-1)/4 \rfloor$ after reduction modulo p. For $n < p$, put $r_0 = p-1-n$. The congruence $\binom{n}{j} \equiv (-1)^j \binom{p-1-j}{r_0} \pmod{p}$, extended by zero for $j > n$, makes its fourth power a polynomial in j of degree $4r_0$. If $4r_0 < p-1$, every monomial sums to zero on $\mathbb{F}_p$; the constant term sums to $p = 0$ as well. Thus $f_n \equiv 0 \pmod{p}$ whenever $n > 3(p-1)/4$.

Lemma I.6.1 proves that $P = B_p h_{\mathrm{can}}$ is a polynomial over $\mathbb{F}_p$, with

$$\deg P \le \deg B_p + p + 1 < 2p < p^2.$$

Lemmas DIGIT and TATE give $h_{\mathrm{can}} - T_p/B_p \equiv 0 \pmod{x^{p^2}}$. Since $B_p(0) = 1$, multiplication by B_p preserves this vanishing. But $\deg(P - T_p) < p^2$, because $\deg T_p \le p-1$. Hence $P - T_p = 0$. This proves $h_{\mathrm{can}} = T_p/B_p$ globally over $\mathbb{F}_p$, and establishes (I.7.1) in all degrees.

I.7.2 Differential identity

Lemma I.4.1 gives $\theta \log q = 1/(G\sqrt{\Delta})$, with square root of constant term one. Differentiate $q(\sigma) = q(x)^p$ in characteristic zero, use (I.7.1), divide by p, and reduce modulo p. The result is

$$1 + \theta h_{\mathrm{can}} = \frac{G(x^p)\sqrt{\Delta(x^p)}}{G(x)\sqrt{\Delta(x)}}.$$

Lucas's congruence gives $G(x) \equiv B_p(x) G(x^p) \pmod{p}$. The normalized square root is integral for odd p, and $\sqrt{\Delta(x^p)}/\sqrt{\Delta(x)} \equiv \Delta(x)^{(p-1)/2} \pmod{p}$. This proves (I.7.2), first as a series at zero and then as an identity of rational functions.

□

I.8 Harmonic collapse and the finite weighted reduction

Lemma I.8.1 (DOUGALL)**.** *Using* H_n, f_n, d_n *as above, put* $e_n = \sum_{j=0}^{n} (1+8j)\binom{n}{j}^4 (H_n - H_j)$. *Then, for every integer* $n \ge 0$,

$$e_n = (4n+1)d_n + (-1)^n \binom{2n}{n} - f_n. \tag{I.8.1}$$

Proof. The established Rogers–Dougall very-well-poised 5F4 sum is NIST DLMF 16.4.9 . Substitution a=b=c=d=-s, and multiplication by s, gives

$$\sum_{j \ge 0} (s - 2j) \binom{s}{j}^4 = \frac{s\Gamma(1+2s)}{\Gamma(1-s)\Gamma(1+s)^3}. \tag{I.8.2}$$

Initially use generic noninteger s with $\operatorname{Re} s > -1/2$; the well-poised factor $(1-s/2)_j/(-s/2)_j = 1 - 2j/s$ gives exactly the summand. Any apparent exceptional hypergeometric parameters are

removed in (I.8.2). Both sides are analytic in a neighborhood of every positive integer n: on a compact such neighborhood the generalized-binomial asymptotic is $O(j^{-\operatorname{Re} s-1})$ uniformly, including the reciprocal $\Gamma(-s)$ factor, so the summands and their s-derivatives are bounded by convergent power series with at most an extra $\log j$. Thus (I.8.2) extends there and can be differentiated termwise. Alternatively the same uniform bounds follow from the gamma quotient for $\binom{s}{j}$ with finitely many initial j removed.

For $j > n$, $\binom{s}{j}^4$ has a fourth-order zero at $s = n$, so its first derivative contributes zero. For $j \le n$,

$$\frac{d}{ds}\binom{s}{j}\bigg|_{s=n} = \binom{n}{j}(H_n - H_{n-j}).$$

The derivative of the right side of (I.8.2) at $n \ge 1$ is $(-1)^n\binom{2n}{n}$, because $1/\Gamma(1-s) = (-1)^n(n-1)!(s-n) + O((s-n)^2)$. Hence

$$f_n + 4\sum_{j=0}^{n}(n-2j)\binom{n}{j}^4(H_n - H_{n-j}) = (-1)^n\binom{2n}{n}.$$

Replace j by n-r. This gives

$$8\sum_{r=0}^{n} r\binom{n}{r}^4(H_n - H_r) - 4nd_n = (-1)^n\binom{2n}{n} - f_n,$$

which is (I.8.1). At n=0 both sides of (I.8.1) vanish directly. This proves its full stated scope.

I.8.1 Exact consequence for the finite weighted sum

For an odd prime p and an integer m prime to p, put $x = 1/m$ and

$$B = \sum_{n<p} f_n x^n, \quad T = \sum_{n<p} 4d_n x^n, \quad R = (1+4\theta)B, \quad C = \sum_{n<p}(-1)^n\binom{2n}{n}x^n,$$

where $\theta = x\,\mathrm{d}/\mathrm{d}x$. Combining (I.8.1) and Lemma I.2.1 gives

$$W_p(m) \equiv m^{p-1}\{-R + p(1+4\theta)T + 4pB + 4pC\} \pmod{p^2}. \tag{I.8.3}$$

Indeed, with the truncations in this finite calculation, $E = \sum_{n<p} e_n x^n = (4\theta+1)T/4 + C - B$, and Lemma I.2.1 gives $W \equiv m^{p-1}(-R + 8pB + 4pE) \pmod{p^2}$. Also

$$C \equiv (1+4x)^{(p-1)/2} \pmod{p}, \tag{I.8.4}$$

by the binomial theorem and $(-1)^n\binom{2n}{n} = 4^n\binom{-1/2}{n}$ for $n < p$. Hence, at $x = 1/36$ and $p \ge 7$, $C(a) \equiv \left(\frac{10}{p}\right) \pmod{p}$.

□

I.9 Regular Franel zeros lift modulo p-squared

Lemma I.9.1 (SSLIFT)**.** *For every odd prime p and every p-adic integer b such that $b\Delta(b)$ is a unit,*

$$B(b) \equiv 0 \pmod{p} \quad\Longrightarrow\quad B(b) \equiv 0 \pmod{p^2}, \qquad T(b) \equiv 0 \pmod{p}. \tag{I.9.1}$$

Every such reduced zero has exact multiplicity two. In particular, if $m = b^{-1}$, Lemma I.2.1 gives $U_p(m) \equiv 0 \pmod{p^2}$. The local indeterminate in the proof below is x; the letter A in its differential invariant is the polynomial specified there and is not a Hasse invariant.

Proof.

I.9.1 The two formal periods and the exact operator

Lemma I.3.1 proves, over rational formal logarithmic series,

$$LF = 0, \qquad L(F\log x + H) = 0,$$

where $h_n = \ell_n = 4d_n$, $F = \sum f_n x^n = G(x)$, $H = \sum h_n x^n$, and $\theta = x\,d/dx$. In this proof only, L denotes the differential operator $\mathcal{L}$ of Lemma I.3.1, while H is its analytic logarithmic companion. Explicitly,

$$L = \theta^3 - 2x(2\theta+1)(3\theta^2+3\theta+1) - 4x^2(\theta+1)(4\theta+3)(4\theta+5). \tag{I.9.2}$$

In particular $LH + L_1F = 0$, where

$$L_1 = 3\theta^2 - 2x(18\theta^2+18\theta+5) - 4x^2(48\theta^2+96\theta+47). \tag{I.9.3}$$

This uses the exact commutator $[L, \log x] = L_1$, not an asymptotic logarithmic relation.

For $n < p$, the numbers f_n, h_n are p-integral. Also

$$f_{p-1} \equiv \sum_{j=0}^{p-1} 1 = 0 \pmod{p}, \qquad v_p(h_p) \geq -1. \tag{I.9.4}$$

The first congruence follows from $\binom{p-1}{j} \equiv (-1)^j$. The second follows directly from the definition: all denominators in H_p have p-adic valuation at most1.

The truncation residuals are supported only at degrees p,p+1:

$$LB = -p^3 f_p x^p - 4p(4p-1)(4p+1)f_{p-1}x^{p+1} \equiv 0 \pmod{p^2}, \tag{I.9.5}$$

and

$$\begin{aligned} LT + L_1B = {} & (-p^3h_p - 3p^2f_p)x^p \\ & - 4\{p(4p-1)(4p+1)h_{p-1} + (48p^2-1)f_{p-1}\}x^{p+1} \equiv \quad 0 \pmod{p}. \end{aligned} \tag{I.9.6}$$

For degrees below p these follow from (I.9.2),(I.9.3). At degree p the displayed missing coefficients give the residuals; at degree p+1 only the quadratic-in-x terms survive. Thus (I.9.4) proves both stated congruences.

I.9.2 Exact differential invariants

For a function Y define the polynomial differential expression

$$\mathcal{C}(Y) = x^2\Delta(2YY'' - Y'^2) + 2xAYY' - 4x(1+15x)Y^2, \quad A = 1 - 18x - 128x^2. \tag{I.9.7}$$

For two functions Y,Z define its symmetric polarization

$$\mathcal{C}_b(Y,Z) = x^2\Delta(YZ'' + ZY'' - Y'Z') + xA(YZ' + ZY') - 4x(1+15x)YZ. \tag{I.9.8}$$

Direct differentiation, using (I.9.2), gives the exact identities

$$\mathcal{C}(Y)' = \frac{2Y}{x}LY, \qquad \mathcal{C}_b(Y,Z)' = \frac{YLZ + ZLY}{x}. \tag{I.9.9}$$

For

$$\mathcal{J}(Y,Z) = \mathcal{C}_b(Y,Z) + \frac{x}{2}(\Delta Y^2)', \tag{I.9.10}$$

one obtains

$$\mathcal{J}(Y,Z)' = \frac{Y(LZ + L_1Y) + ZLY - \mathcal{C}(Y)}{x}. \tag{I.9.11}$$

For checking (I.9.11), the remaining elementary identity is

$$\left[\frac{x}{2}(\Delta Y^2)'\right]' = \frac{YL_1Y - \mathcal{C}(Y)}{x}.$$

These are formal polynomial differential identities after multiplication by x. The named checking script reconstructs them with arbitrary symbolic jets; their exact formulas, not finite prime testing, establish their uniform scope.

Since F has constant term1 and H has constant term0, (I.9.9),(I.9.11),(I.9.2),(I.9.3) imply

$$\mathcal{C}(F) = 0, \qquad \mathcal{J}(F,H) = 0 \tag{I.9.12}$$

as rational formal power series: their derivatives vanish in characteristic0 and their constant terms vanish.

I.9.3 The invariants survive truncation

The polynomial $\mathcal{C}(B)$ has degree at most2p and constant term0. From (I.9.5),(I.9.9), its derivative is0 modulo p^2. Every coefficient at a degree not divisible by p is therefore0 modulo p^2. Only the coefficients at p and2p require separate checks.

The coefficient at p in $\mathcal{C}(F) - \mathcal{C}(B)$ is exactly $2p^2 f_p$: the two contributions involving the omitted term $f_p x^p$ and the constant term1 are $2p(p-1)f_p$ and $2pf_p$. All remaining omitted terms first affect larger degrees. By (I.9.12), the coefficient at p of $\mathcal{C}(B)$ is consequently $-2p^2 f_p$. The coefficient at2p is an integer multiple of f_{p-1}^2, since it uses only the two leading terms of B. By (I.9.4) it too vanishes modulo p^2. We conclude

$$\boxed{\mathcal{C}(B) \equiv 0 \pmod{p^2}.} \tag{I.9.13}$$

Similarly $\mathcal{J}(B,T)$ has degree at most2p, constant term0, and by (I.9.5),(I.9.6),(I.9.11),(I.9.13), derivative0 modulo p. Hence only its coefficients at p,2p can survive modulo p.

At p the difference $\mathcal{J}(F,H) - \mathcal{J}(B,T)$ has coefficient $p^2 h_p + p f_p$. The h_p term contributes $p(p-1)h_p + ph_p$ in (I.9.8); the f_p term contributes pf_p through the last term of (I.9.10). There is no further contribution from (I.9.8) involving f_p since H has constant term0. By (I.9.4) this coefficient is0 modulo p. At2p every term has a factor $f_{p-1}h_{p-1}$ or f_{p-1}^2, again0 modulo p. Thus

$$\boxed{\mathcal{J}(B,T) \equiv 0 \pmod{p}.} \tag{I.9.14}$$

I.9.4 Evaluate at a nonsingular zero

Let $\bar{a}$ be a zero of B over an algebraic closure of $\mathbb{F}_p$ with $\bar{a}\Delta(\bar{a}) \neq 0$, and let its multiplicity be r. Since B(0)=1 and deg B<=p-1, $1 \leq r \leq p-1$. In (I.9.13) modulo p, the lowest potentially nonzero coefficient at this root comes from the first term of (I.9.7); after writing $B = (x-\bar{a})^r u$, it is

$$\bar{a}^2\Delta(\bar{a})\, r(r-2)u(\bar{a})^2(x-\bar{a})^{2r-2}.$$

The other terms have order at least $2r-1$. Thus $r(r-2) = 0$ in $\mathbb{F}_p$, forcing $r = 2$. At each such root, B' vanishes and B'' is nonzero.

At a p-adic lift a, both $B(a), B'(a)$ are divisible by p, and $B''(a)$ is a unit. By (I.9.13), all terms of (I.9.7) except $2a^2\Delta(a)B(a)B''(a)$ vanish modulo p^2. Its other factors are units, so $B(a) \equiv 0 \pmod{p^2}$.

By (I.9.14) at the reduced root, all terms of (I.9.10) vanish except $\bar{a}^2\Delta(\bar{a})T(\bar{a})B''(\bar{a})$. Hence T(a)=0 modulo p. This proves the zero-lifting assertion.

Finally, Lemma I.2.1 with $a = m^{-1}$ gives

$$\sum_{n<p} S_n(m) \equiv m^{p-1}\{B(a) - pT(a)\} \pmod{p^2}.$$

Both terms in braces vanish modulo p^2 by the zero-lifting assertion, proving the final assertion. □

I.10 Exact CM value, derivative normalization, and reduction type

Lemma I.10.1 (CMPOINT)**.** *Put $\alpha = \mathrm{i}\sqrt{3/10}$ and $a = 1/36$. Then $x(\alpha) = a$, $(1+4\theta)G(a) = 18/(\pi\sqrt{15})$, and*

$$\frac{x(\alpha)}{x'(\alpha)}\left(\frac{2}{\alpha-\bar{\alpha}} + \frac{\Theta'(\alpha)}{\Theta(\alpha)}\right) = -\frac{1}{4}.$$

At every prime $p > 5$, the CM curve has potentially good reduction, and $B_p(a) \equiv 0 \pmod{p}$ if and only if $\left(\frac{-30}{p}\right) = -1$. In that case $U_p(36) \equiv 0 \pmod{p^2}$.

Proof.

I.10.1 Exact CM value, with rigorous root selection

Let

$$X(\tau) = \frac{\eta(2\tau)^6\eta(5\tau)^6}{\eta(\tau)^6\eta(10\tau)^6}, \quad x(\tau) = \frac{X}{(X-1)^2}, \quad \alpha = i\sqrt{3/10}, \quad a = 1/36.$$

Then

$$\boxed{x(\alpha) = a.} \tag{I.10.1}$$

The primary source is S. Cooper, *Level 10 analogues of Ramanujan's series for 1/pi*, J. Ramanujan Math. Soc. **27** (2012), 59–76, author manuscript dated March 25, 2011, author-uploaded full text, definitions (11), (15), and the proof of Theorem 5.2, printed pp. 13–14, equation (62). In that source $x_C = 1/X$ and $y_A = x_C/(1-x_C)^2 = x$, so the parameter agrees exactly. Equation (62) states that, for $u = x(\tau)$ and $v = x(3tau)$,

$$\begin{aligned} P(u,v) = {} & u^4 + v^4 - uv + 12(u^2v + uv^2) + 18(u^3v + uv^3) \\ & - 48(u^4v + uv^4) + 387u^2v^2 + 288(u^3v^2 + u^2v^3) \\ & + 864(u^4v^2 + u^2v^4) - 9568u^3v^3 \\ & - 6912(u^4v^3 + u^3v^4) + 20736u^4v^4 = 0. \end{aligned} \tag{I.10.2}$$

This established modular equation, not a fitted numerical relation, is the source dependency.

Eta inversion gives $X(-1/(10tau)) = X(\tau)$ exactly: the eta factors acquire powers `(5,2)` in the numerator and `(10,1)` in the denominator, whose scalar products agree. Hence x is invariant under this Fricke transformation. Set $\beta = i/\sqrt{30}$, so $3beta = \alpha = -1/(10beta)$. Then $u = x(\beta) = x(\alpha) = v$ in (I.10.2). Exact substitution factors it as

$$P(v,v) = v^2(v-1)(4v-1)(4v+1)^2(9v+1)(36v-1). \tag{I.10.3}$$

Here is a rigorous replacement for the source's numerical root selection. For real $0 < Q < 1$, eta products give

$$X = Q^{-1}\prod_{5\nmid m}(1+Q^m)^6 > Q^{-1}.$$

At $\tau = \alpha$, we have $Q = \exp(-2pi\sqrt{3/10}) < \exp(-3) < 1/18$. The first inequality follows from $\pi > 3$ and $\sqrt{3/10} > 1/2$; the second follows, for example, from the exponential Taylor sum through degree five, which is $92/5 > 18$. Thus $X(\alpha) > 18$, and since $X/(X-1)^2$ is positive and decreasing for $X > 1$,

$$0 < x(\alpha) < 18/17^2 < 1/16.$$

Among the roots in (I.10.3), precisely $1/36$ belongs to this interval. This proves (I.10.1).

The named exact script `algebra_cm36_check.py` verifies (I.10.3) and the derivative computations below, using SymPy over the rationals. Its certificate is `algebra_cm36_certificate.json`, with output `algebra_cm36_check.txt`. Reproduce from this part's directory with `pythonalgebra_cm36_check.py`, intended timeout 30 seconds; Python 3.13.6, SymPy 1.14.0. This finite algebra check supports the explicit factorization; modularity of (I.10.2) is supplied by the cited established source.

I.10.2 Exact weighted normalization

Put

$$\Theta(\tau) = \frac{10E_2(10\tau) + 5E_2(5\tau) - 2E_2(2\tau) - E_2(\tau)}{12}, \quad \Delta(x) = (1-16x)(1+4x).$$

Cooper's established identities are $\Theta = G(x)$ and

$$\frac{dx}{d\tau} = 2\pi i\, x\Theta\sqrt{\Delta(x)}, \tag{I.10.4}$$

with the root normalized at the cusp; see the same source (23), (33), (47). The established Yang series is

$$(1+4\theta)G(1/36) = \frac{18}{\pi\sqrt{15}}, \qquad \theta = x\frac{d}{dx}. \tag{I.10.5}$$

It is (2) in that paper and the $N = 3$ case of its Theorem 5.2. Convergence at $1/36$ is absolute, including the differentiated series, since $f_n \le 16^n$.

Define exactly the quantity in Beukers' weighted CM theorem,

$$\delta(\alpha) = \frac{x(\alpha)}{x'(\alpha)} \left(\frac{2}{\alpha - \bar{\alpha}} + \frac{\Theta'(\alpha)}{\Theta(\alpha)} \right).$$

Then

$$\boxed{\delta(\alpha) = -1/4.} \tag{I.10.6}$$

Indeed $\Delta(a) = 50/81$, with positive square root `5sqrt(2)/9` on the real cusp branch up to α, and $Im(\alpha) * sqrt(\Delta(a)) = \sqrt{15}/9$. Equations (I.10.4)–(I.10.5) give

$$\begin{aligned} \delta(\alpha) &= \frac{\theta G(a)}{G(a)} - \frac{1}{2\pi \ \mathrm{Im}(\alpha) G(a) \sqrt{\Delta(a)}} \\ &= \left(\frac{9}{2\pi\sqrt{15} G(a)} - \frac{1}{4} \right) - \frac{9}{2\pi\sqrt{15} G(a)} = -\frac{1}{4}. \end{aligned}$$

The identity $\Theta = G(x)$ holds along this branch because x stays in $(0, 1/16)$ up to α, as the same elementary $X > 18$ bound shows. The square-root sign stays positive since Δ has no zero on that interval. Thus the weight `1-n/delta` is exactly `4n+1`; there is no numerical or unspecified modular normalization left in this step.

I.10.3 CM order and good reduction hypotheses

Let $E_\alpha = \mathbb{C}/(\mathbb{Z} + \alpha\mathbb{Z})$, with the standard cyclic level10 structure. Its full complex endomorphism ring is

$$\mathrm{End}(E_\alpha) = \mathbb{Z}[\sqrt{-30}] = \mathcal{O}_{\mathbb{Q}(\sqrt{-30})}. \tag{I.10.7}$$

An endomorphism is multiplication by $a + b\alpha$ with $a, b \in \mathbb{Z}$, since it sends one into the lattice. As $\alpha^2 = -3/10$, preserving α forces $10 \mid b$. Conversely these multipliers preserve the lattice. Since $-30 \equiv 2 \pmod 4$, this is the maximal order, of discriminant -120.

The CM elliptic curve, its endomorphisms, and its finite level structure are defined over a number field after a finite extension. A CM elliptic curve has potential good reduction at every finite prime. This is the established theorem in J.S. Milne, *Modular Functions and Modular Forms*, author notes, Theorem12.14, printed page125; its proof invokes the CM Tate-module action and Néron–Ogg–Shafarevich. Corollaries12.15–12.16 also give algebraicity/integrality of the CM j-invariant. Thus for each fixed $p > 5$, enlarge the field to obtain good reduction of E_alpha. Its prime-to-p cyclic level10 structure extends over that good model. Adding full auxiliary level3 after a further finite extension gives a fine, smooth modular-curve point in characteristic p. Only primes $p \geq 7$ are used, so the auxiliary level and all level10 factors are invertible.

There is no cusp or parameter pole at this specialization. On the interior good-level moduli problem X is an integral modular unit: its fourth power is the ratio $\Delta_2\Delta_5/(\Delta_1\Delta_{10})$ of the discriminant forms of the prime-to-p quotient elliptic curves, up to differential factors that are p-adic units. All four discriminants are units on smooth elliptic curves. Moreover (I.10.1) gives exactly

$$X(\alpha)^2 - 38X(\alpha) + 1 = 0, \qquad (X(\alpha) - 1)^2 = 36X(\alpha). \tag{I.10.8}$$

The first equation makes $X(\alpha)$ an algebraic unit, and the second makes $X(\alpha) - 1$ a p-adic unit for every $p > 5$. Hence $x = X/(X-1)^2$ is regular at the reduced CM point and reduces to $a = 1/36 \in \mathbb{F}_p$. This local unit argument justifies the specialization through the quotient.

Deuring's reduction theorem now applies: an elliptic curve with CM by an order in an imaginary quadratic field K, at a place of good reduction, is supersingular exactly when the rational prime is inert or ramified in K, and ordinary exactly when it splits. A primary author source with the exact statement is A. Zaytsev, *Generalization of Deuring reduction theorem*, arXiv:1209.5207, Theorem1.1, printed page1; §5.1 also reconstructs the elliptic case. It cites Lang, *Elliptic Functions*, second edition, Theorem12, page182. Here $p > 5$ does not divide120, so it is unramified; by (I.10.7) the two alternatives are precisely $(-30/p) = -1$ and$+1$. Good reduction and the order have both been checked above.

I.10.4 Hasse detection of the CM reduction and the inert case

Lemma I.6.1 proves on a fine auxiliary-level cover the exact equal-weight Hasse identity

$$B_p(x) = A_{\mathrm{Hasse}}^2/\Theta^{p-1}. \tag{I.10.9}$$

At a finite supersingular point, the local order calculation in that lemma forces Θ to be a unit and the pullback of $B_p(x)$ to have order two; at a finite ordinary point, both are units. All its hypotheses hold at the reduced CM point, by the good-reduction and local-unit argument just proved. Deuring's theorem therefore gives, for every $p > 5$,

$$B_p(a) = 0 \pmod p \quad \Longleftrightarrow \quad (-30/p) = -1. \tag{I.10.10}$$

Since $a\Delta(a) = \frac{1}{36}\frac{50}{81}$ is a unit for these primes, Lemma I.9.1 and Lemma I.2.1 imply

$$(-30/p) = -1 \quad \Longrightarrow \quad U_p(36) \equiv 0 \pmod{p^2}. \tag{I.10.11}$$

The normalized degree-three correspondence and its source-coordinate derivatives are proved in Lemma I.13.1 below. In that lemma $\beta = i/\sqrt{30}$, $u = x(\tau)$, $v = x(3\tau)$, and $M_3 = \Theta(\tau)/\Theta(3\tau)$ satisfy exactly

$$u(\beta) = v(\beta) = a, \quad v'(a) = -1, \quad v''(a) = 144/5, \quad M_3(a) = 3, \quad M_3'(a) = -54, \tag{I.10.12}$$

where these primes denote derivatives with respect to the source u. These will also be available as precise normalization facts for subsequent chapters; no arithmetic weighted congruence is inferred from the complex Ramanujan value alone.

□

I.11 Integral canonical period ratio on the ordinary locus

Lemma I.11.1 (CMRATIO)**.** *For every $p \geq 7$, the normalized canonical weight-two period ratio ρ is a regular unit on the finite ordinary x-locus and satisfies $\rho \equiv B \pmod{p^2}$, including after differentiation in x. The construction is given below.*

Proof.

I.11.1 Atkin-Lehner character, with its normalization

Let $\Theta = G(x(Q))$ and $Y = \eta_5^2\eta_{10}^2/(\eta_1^2\eta_2^2)$, so that $\Theta = (Q\,\mathrm{d}/\mathrm{d}Q)\log Y$. By Lemma I.4.1, Y is a modular unit with trivial character and cusp orders $(-1,-1,1,1)$ at denominators $1, 2, 5, 10$. The operation w_e toggles the prime divisors of e in the cusp denominator. Divisor comparison on the compact genus-zero curve shows that $Y \circ w_2$ is a nonzero constant times Y, whereas $Y \circ w_5$ is a constant divided by Y. Logarithmic differentiation gives

$$\Theta|_2 w_2 = \Theta, \quad \Theta|_2 w_5 = -\Theta, \qquad \chi(w_1) = \chi(w_2) = 1, \quad \chi(w_5) = \chi(w_{10}) = -1. \tag{I.11.1}$$

Here $\Theta|_2\gamma = (\det\gamma)(c\tau+d)^{-2}\Theta(\gamma\tau)$. Thus, if $\gamma \in W_e$ has determinant e,

$$\Theta(\gamma\tau) = \frac{\chi(w_e)}{e}(c\tau+d)^2\Theta(\tau). \tag{I.11.2}$$

The factor 1/e follows from the determinant-normalized slash convention.

I.11.2 All-index congruence and the cusp ratio

Use the established all-index Gessel-Lucas theorem of A. Straub, author version dated December10,2024, Theorems1.3/5.1 and Section4: author PDF and author corrigendum. Precisely, for every $a \geq 0,\ 0 \leq b < p$,

$$f_{pa+b} \equiv f_a(f_b + pa\ell_b) \pmod{p^2}. \tag{I.11.3}$$

Its s10 sequence is exactly fourth Franel, and the formal derivative ell_b is the harmonic coefficient here: the differentiated recurrence and initial values agree with Lemma I.3.1 (alternatively differentiate the generalized-binomial fourth-power series at a nonnegative integer; all omitted terms have order four, and binomial symmetry identifies the two harmonic expressions). The theorem's lower digit has the range $0 \leq b < p$; the higher index a is unrestricted.

Sum (I.11.3) coefficientwise, with $\theta = x\,\mathrm{d}/\mathrm{d}x$, to obtain

$$G(x) \equiv B(x)G(x^p) + pT(x)(\theta G)(x^p) \pmod{p^2}.$$

Lemma I.7.1 identifies $\sigma \equiv x^p(1 + pT/B) \pmod{p^2}$. Taylor expansion of the integral series G therefore gives

$$\frac{G(x)}{G(\sigma(x))} \equiv B(x) \pmod{p^2}. \tag{I.11.4}$$

as formal series at zero. The next construction transports this cusp identity into the integral ordinary function ring.

I.11.3 Why the cusp congruence specializes on the ordinary locus

Use the smooth integral level-ten model for $p \geq 7$, adding an auxiliary fine level prime to p if needed, and remove the cusps and $x = \infty$. On the ordinary locus the canonical connected p-subgroup defines σ, commuting with prime-to-p Atkin–Lehner operations as proved in Lemma I.6.1. For a local invariant differential ω, a quotient generator satisfies $\phi^*\omega' = pu\omega$, with u a unit. Indeed the connected ordinary p-isogeny has differential p times a unit and its dual has unit differential. On $\mathrm{Tate}(Q) \to \mathrm{Tate}(Q^p)$, we have $u = 1$.

Define ρ as $p^2\Theta$ on the source divided by the isogeny's pullback of Θ on the target. In these generators it is $\Theta(\text{source})/(u^2\Theta(\text{target}))$. It is independent of generators and is a regular unit when both source and target have finite ordinary x. The Hasse identity $B = A_{\mathrm{Hasse}}^2/\Theta^{p-1}$ from Lemma I.6.1 makes Θ a unit there. The Atkin–Lehner characters cancel and the canonical quotient commutes with those operations, so ρ descends to the ordinary x-line. Its cusp expansion is $\Theta(Q)/\Theta(Q^p) = G(x)/G(\sigma(x))$.

Therefore (I.11.4) is a congruence of regular functions on the ordinary formal x-line, not merely a formal specialization of G. To justify passage from the expansion: reduce the difference rho-B modulo p, viewed as a rational function on the irreducible ordinary x-line;

its zero expansion forces that function to be zero. Divide the regular difference by p and repeat modulo p to obtain zero modulo p^2. Concretely, adjoining the ordinary cusp by its Tate extension gives the p-adic completion of the localized coordinate ring $\mathbb{Z}_p[x, 1/B]$, with Delta inverted if desired. The canonical quotient extends there on Tate(Q), and its divided differential is the identity at that cusp. Reduction of a regular formal function is a rational function on the irreducible ordinary x-line; the expansion of such a function at the cusp is injective. The good model is p-flat, so division by p after a zero reduction is valid locally and the local divisions glue. Repeating this argument proves the modulus p^2. The derivation d/dx preserves this completed localized ring and its ideal p^2. The same congruence holds after differentiation in the etale finite coordinate x. Thus

$$\rho(a) \equiv B(a), \qquad (\theta\rho)(a) \equiv (\theta B)(a) \pmod{p^2}. \tag{I.11.5}$$

at every ordinary lift a with the stated finite-coordinate condition.

□

I.12 The four split CM branches and the ordinary weighted value

Lemma I.12.1 (SPLIT). *For every prime $p > 5$ with $\left(\frac{-30}{p}\right) = 1$, the four representation cases of* (1.1) *hold, and $R_p(a) \equiv p\left(\frac{-15}{p}\right) \pmod{p^2}$.*

Proof.

I.12.1 CM root and its unique ordinary branch

Put $\alpha = \sqrt{-30}/10$, of positive imaginary part, and $a = x(\alpha) = 1/36$, as established in Lemma I.10.1. The elliptic curve $\mathbb{C}/(\mathbb{Z}+\mathbb{Z}\alpha)$ has endomorphism ring $\mathbb{Z}[\sqrt{-30}]$ of discriminant -120, and its cyclic order-ten subgroup is generated by 1/10. It has good reduction after finite extension at every $p > 5$. Its level-ten point extends on the proper good model and cannot reduce to a cusp: all cusp images under x are zero, whereas $a \bmod p \neq 0$. A fine cover supplies a smooth elliptic scheme there.

Suppose integers c,d and e in {1,2,5,10} satisfy 10|c, e|d, and

$$ep = d^2 + \frac{3}{10}c^2 = |c\alpha + d|^2.$$

Write $\pi = c\alpha + d$, and choose the embedding $K \hookrightarrow \mathbb{Q}_p$ for which π is a p-adic unit. This is possible because $\pi\bar{\pi} = ep$: divisibility of both conjugates by p would force $p \mid c, d$ and hence $p^2 \mid ep$. The endomorphism π has unit differential and degree ep; its reduction has an etale p-kernel, so the CM reduction is ordinary.

Let $c' = -c$, $d' = d$. We check $\gcd(c'/e, d') = 1$, including primes dividing ten. Write $c = 10c_0$, $d = ed_0$, so $p = ed_0^2 + (30/e)c_0^2$. A common prime outside ten divides d_0, c_0, forcing its square to divide p, impossible. A common factor two forces d_0 even when e is odd, or c_0 even when e is even; either makes p even. A common factor five forces $5 \mid d_0$ when $5 \nmid e$, or $5 \mid c_0$ when $5 \mid e$; either gives $5 \mid p$. Choose $a_0, b \in \mathbb{Z}$ with $a_0 d' - b(c'/e) = 1$, and put $A = ea_0$. Then $\gamma = \left(\begin{smallmatrix} A & b \\ c' & d' \end{smallmatrix}\right)$ has determinant e, satisfies $10 \mid c'$ and $e \mid A, d'$, and belongs to W_e. Exact multiplication using $\alpha^2 = -3/10$ gives

$$\gamma\alpha = \frac{\alpha + k}{p}, \qquad k = \frac{A}{e}c'\frac{3}{10} + \frac{bd'}{e} \in \mathbb{Z}. \tag{I.12.1}$$

The branch $\tau \mapsto (\tau + k)/p$ therefore returns α to the same Atkin–Lehner orbit. By (I.12.2), its normalized ratio at α is

$$\rho_k(\alpha) = \frac{p^2\Theta(\alpha)}{\Theta((\alpha+k)/p)} = \frac{\chi(w_e)}{e}\pi^2. \tag{I.12.2}$$

This is a p-adic unit. An ordinary elliptic curve has exactly one connected p-subgroup, whose normalized ratio is a unit. The other p-isogeny subgroups have unit differential, so their p^2-normalized ratios are divisible by p^2 when the reduced source and target have finite ordinary coordinate. Here those coordinates reduce to a and its Frobenius or inverse-Frobenius images, all equal to $a \in \mathbb{F}_p$. Thus this is the canonical branch used in (I.11.5). In particular $\sigma(a) = a$, and that equation gives

$$B(a) \equiv \frac{\chi(w_e)}{e}\pi^2 \pmod{p^2}. \tag{I.12.3}$$

At this fixed point Lemma I.7.1 gives $T(a) \equiv q_p(36)B(a) \pmod{p}$, and Lemma I.2.1 gives $U_p(36) \equiv B(a) \pmod{p^2}$. The conjugate of (I.12.2) is divisible by p^2, so

$$U_p(36) \equiv \chi(w_e)\left(\frac{4d^2}{e} - 2p\right) \pmod{p^2}. \tag{I.12.4}$$

I.12.2 Four representations and exact target branches

Reduction of primitive positive definite forms of discriminant -120 gives precisely $(1,0,30), (2,0,15), (3,0,10), (5,0,6)$. Indeed, enumerate $A \le \sqrt{40} < 7$, $|B| \le A$, and $C = (B^2+120)/(4A) \ge A$, with the usual boundary sign convention. Every split prime $p > 5$ is represented: choose $r \in \mathbb{Z}$ with $r^2 \equiv -30 \pmod{p}$. The primitive positive definite integral form $(p, 2r, (r^2+30)/p)$ represents p at $(1,0)$, and integral unimodular reduction puts it in the displayed list. Primitivity follows from $p \nmid r$. The only form-theoretic input is the elementary reduction theorem; see Cox, *Primes of the Form* $x^2 + ny^2$, §2.

The sign classification follows from reduction modulo three and five. For the forms $(1,0,30), (2,0,15), (3,0,10), (5,0,6)$, the pairs $\big((p/3),(p/5)\big)$ are, respectively, $(1,1), (-1,1), (-1,-1), (1,-1)$; check the nonzero square in each reduced equation, using $p > 5$. Quadratic reciprocity gives $(-30/p) = (2/p)(p/3)(p/5)$, determining the remaining sign. These triples are exactly the target cases; no genus-number or class-group table is needed.

For the forms in the target order choose

form	e	c	d	$\chi(w_e)$
$p = x_0^2 + 30y_0^2$	1	$10y_0$	x_0	1
$p = 3x_0^2 + 10y_0^2$	10	$10x_0$	$10y_0$	-1
$p = 2x_0^2 + 15y_0^2$	2	$10y_0$	$2x_0$	1
$p = 5x_0^2 + 6y_0^2$	5	$10y_0$	$5x_0$	-1

The norm equation and level divisibilities hold in every row. Substituting into (I.12.4) gives exactly the first, second, third and fourth branches of target I, respectively. Thus no split genus or sign is omitted.

I.12.3 Weighted ordinary congruence

Lemma I.10.1, equation (I.10.6), gives $\delta(\alpha) = -1/4$, where

$$\delta(\alpha) = \frac{a}{x'(\alpha)} \left(\frac{2}{\alpha - \bar{\alpha}} + \frac{\Theta'(\alpha)}{\Theta(\alpha)} \right).$$

Differentiate $\rho_k(\tau) = p^2\Theta(\tau)/\Theta((\tau+k)/p)$ at alpha, using (I.11.2) and (I.12.1). Ordinary differentiation gives

$$(\theta\rho)(a) = \frac{\chi(w_e)}{e} c\pi(\alpha - \bar{\alpha})\delta(\alpha). \tag{I.12.5}$$

For clarity, differentiating the transformation law gives

$$e\frac{\Theta'(\gamma\tau)}{\Theta(\gamma\tau)} = 2c'(c'\tau + d') + (c'\tau + d')^2 \frac{\Theta'(\tau)}{\Theta(\tau)}.$$

Substitute $\tau = \alpha$ into the derivative of ρ_k, use $\pi\bar{\pi} = ep$ and $c'\alpha + d' = \bar{\pi}$, and simplify. This gives (I.12.5) without truncating an analytic infinite sum at a p-adic unit. Equation (I.11.5) permits replacing $\rho, \theta\rho$ by $B, \theta B$ modulo p^2. Hence

$$(1+4\theta)B(a) \equiv \rho(a) - \frac{(\theta\rho)(a)}{\delta(\alpha)} = \frac{\chi(w_e)}{e}\{\pi^2 - c\pi(\alpha - \bar{\alpha})\} = \chi(w_e)p \pmod{p^2}.$$

In the four ordinary classes, (I.11.1) gives $\chi(w_e) = \left(\frac{2}{p}\right) = \left(\frac{-15}{p}\right)$. This is the reciprocal weighted value for every ordinary prime $p > 5$.

□

I.13 The inert reciprocal weighted value

Lemma I.13.1 (IRR)**.** *For every $p > 5$ such that $(-30/p) = -1$,*

$$R_p(a) = (1+4\theta)B_p(a) \equiv p(-15/p) \pmod{p^2}. \tag{I.13.1}$$

Proof.

We use Lemma I.10.1 and Lemma I.9.1, which give an exact double reduced zero and $B(a) = 0 \pmod{p^2}$, $T(a) = 0 \pmod{p}$; Lemma I.7.1, which gives the global rational identity for $\theta(T/B)$; and Lemma I.11.1, whose ordinary-locus ratio congruence is available before any CM specialization. A regular cross-multiple will be formed on that locus and extended before taking a supersingular value.

I.13.1 The degree-three correspondence and its CM germ

Lemma I.13.2 (Degree-three CM germ)**.** *On the normalized cyclic degree-three correspondence, with $u = x(\tau)$, $v = x(3\tau)$, and $M = \Theta(\tau)/\Theta(3\tau)$, the characteristic-zero point $u = v = a = 1/36$ is rational and has a unique branch. Its source-coordinate derivatives are*

$$v'(a) = -1, \qquad v''(a) = 144/5, \qquad M(a) = 3, \qquad M'(a) = -54.$$

Proof. Let x be precisely the parameter of Lemma I.4.1 and Lemma I.10.1,

$$X = \eta_2^6\eta_5^6/(\eta_1^6\eta_{10}^6), \quad x = X/(X-1)^2, \quad \Theta = (10E_2(10\tau) + 5E_2(5\tau) - 2E_2(2\tau) - E_2(\tau))/12.$$

Thus $\Theta = G(x)$ is precisely the corrected weight-two form, with the same normalization.

On the cyclic3-isogeny correspondence put

$$u = x(\tau), \quad v = x(3\tau), \quad M = \Theta(\tau)/\Theta(3\tau). \tag{I.13.2}$$

Lemma I.4.1 consequently gives the exact cusp identity $G(u) = MG(v)$, as a formal identity on this cyclic3-isogeny correspondence. Let C be the normalized compact curve obtained from $X_0(30)$ by the simultaneous operations w_2, w_5. The ratio M descends to it: intrinsically, $M = 9\Theta(E)/\phi_3^*\Theta(E/C_3)$, with the factor nine normalizing its Tate expansion as in (I.13.2). Prime-to-three Atkin–Lehner isogenies commute with ϕ_3, and their source and target characters and differential factors cancel. The rational Fourier expansion makes M a function over $\mathbb{Q}$.

For an explicit check, take $w_2 = \left(\begin{smallmatrix} 16 & 1 \\ 30 & 2 \end{smallmatrix}\right)$ and $w_5 = \left(\begin{smallmatrix} 25 & 4 \\ 30 & 5 \end{smallmatrix}\right)$ at level thirty. Their determinants are two and five. On the target parameter 3τ, the induced representatives are $\left(\begin{smallmatrix} 16 & 3 \\ 10 & 2 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 25 & 12 \\ 10 & 5 \end{smallmatrix}\right)$, in the same level-ten cosets. Their denominator factors agree with the source factors $30\tau + 2$ and $30\tau + 5$. The weight-two factors and characters therefore cancel in M.

The map C→(u,v) is generically one-to-one. Indeed the four cyclic3 subgroups of a generic elliptic curve give four distinct targets even after the level10 Atkin–Lehner quotient. If two were identified by w_e, $e \in \{2, 5, 10\}$, composing the two degree3 isogenies and the degree-e identification would give an endomorphism of degree $9e$ of an elliptic curve with End $= \mathbb{Z}$, impossible since$9e$ is not a square. For $e = 1$ the standard dual-isogeny cancellation forces the original3-kernels to coincide. Hence $\mathbb{Q}(C) = \mathbb{Q}(u, v)$ and Cooper's degree-four-in-v modular equation is its plane image.

Set $\beta = i/\sqrt{30}$ and $\alpha = 3\beta = -1/(10\beta)$. Equation (I.10.1) gives $u(\beta) = v(\beta) = a$. At (a, a), Cooper's plane equation P satisfies

$$P_u = P_v = 4375/236196 \neq 0 \quad \text{over Q}. \tag{I.13.3}$$

This is a smooth point of the characteristic-zero plane curve, with a unique point P of its normalization above it. Its residue field is $\mathbb{Q}$, so $P \in C(\mathbb{Q})$. This characteristic-zero assertion does not infer uniqueness from the reduced plane equation at seven.

The exact derivatives of its characteristic-zero germ are

$$v(a) = a, \qquad v'(a) = -1, \qquad v''(a) = 144/5, \qquad M(a) = 3, \qquad M'(a) = -54, \tag{I.13.4}$$

Derivatives are with respect to the source coordinate u. Implicit differentiation of P, checked by the exact certificate, gives v', v''. The weight-two Fricke law $\Theta(-1/(10\tau)) = -10\tau^2\Theta(\tau)$, evaluated at β, gives $\Theta(\alpha) = \Theta(\beta)/3$, hence $M(a) = 3$.

For the derivative, Cooper's modular differential identity gives

$$M = 3\frac{v}{u}\frac{\sqrt{\Delta(v)}}{\sqrt{\Delta(u)}}\frac{1}{v'}.$$

The square-root branches have opposite signs at β, consistently with $M(a) = 3$ and $v'(a) = -1$. Their constant relative sign disappears on logarithmic differentiation, giving

$$\frac{M'(a)}{M(a)} = -\frac{2}{a} - \frac{\Delta'(a)}{\Delta(a)} - \frac{v''(a)}{v'(a)} = -72 + 126/5 + 144/5 = -18.$$

Thus $M'(a) = -54$, with differentiation in the source coordinate u; differentiation in the target coordinate would give $+54$.

□

I.13.2 Integral branch and Frobenius compatibility, including the prime seven

Fix an inert prime $p > 5$. Use the proper good-level model of C over $\mathbb{Z}_p$, where $p \nmid 30$, and its normal tame Atkin–Lehner quotient. Properness extends P to a $\mathbb{Z}_p$-point, whose reduction is $\mathbb{F}_p$-rational. Both CM curves have potentially good reduction and their prime-to-p levels extend, as checked in Lemma I.10.1. Their reductions are interior supersingular points. The local unit identity $(X-1)^2 = 36X$ from that lemma shows that u, v are regular there and equal to a.

We prove that u is etale at the normalized reduced point without dividing by the reduced plane partial derivative. Choose an auxiliary full level $N \geq 3$ coprime to $30p$, for instance seven if $p \neq 7$ and eleven otherwise. The moduli problem is fine and the degree-three isogeny transports this level. On the fine source curve, the Hasse identity gives pullback order two for $B(u)$ at the supersingular point. Lemma I.9.1 gives exact multiplicity two for B as a polynomial in u, since $a\Delta(a) \neq 0$. Thus a ramification index e satisfies $2 = 2e$, so $e = 1$. Adding or forgetting a cyclic subgroup of order three is finite etale, since $E[3]$ is finite etale and its four cyclic subgroups form a finite etale scheme. The composite source-coordinate map is therefore etale. Ramification indices multiply under the tame normalized quotient, giving index one on that quotient as well. Hence $u : C \to \mathbb{P}^1$ is etale at the reduced point.

Consequently the completed local ring of C at P modp is

$$\mathcal{R} = \mathbb{Z}_p[[u - a]]. \tag{I.13.5}$$

The germs $v(u), M(u)$ lie in this ring. Moreover M is a unit: the two Θ sections are units at their finite supersingular points by the Hasse argument of Lemma I.10.1, and the degree-three isogeny's differential is a unit since $p \neq 3$. The residue point is $\mathbb{F}_p$-rational because $P \in C(\mathbb{Q})$, so Frobenius preserves the normalized germ. On an algebraic etale chart over the source coordinate, formal etale lifting uniqueness determines a Frobenius lift from its base-coordinate lift and reduction.

This gives integral local germs at every inert $p > 5$. At seven, where the displayed plane partial derivatives in (I.13.3) vanish on reduction, the argument uses the normalized etale modular correspondence and does not divide by those derivatives.

Let Φ_0 be the etale lift with $\Phi_0(u) = u^p$. It reduces to Frobenius modulo p. In the notation of (I.13.5), $\Phi_0(M) = M(u^p)$, with derivative $pu^{p-1}\Phi_0(M')$. On the ordinary localization the canonical lift σ_C satisfies

$$\sigma_C(M) = \Phi_0(M) + pu^p \frac{T}{B}\,\Phi_0(M') \pmod{p^2}. \tag{I.13.6}$$

For precision, construct these maps on an algebraic etale chart $\mathcal{A}/p^2\mathcal{A}$ before completion or localization. The two maps σ_C and Φ_0, into its completed ordinary localization, have the same reduction, namely absolute Frobenius. Their difference divided by p is a Frobenius-semilinear derivation modulo p. Etaleness over the source coordinate u makes this derivation determined by its value on u, which is u^pT/B. Hence for every regular algebraic germ f,

$$\sigma_C(f) - \Phi_0(f) = pu^p(T/B)\Phi_0(df/du) \pmod{p^2}.$$

Taking $f = M$ proves (I.13.6). It requires no value of T/B at the supersingular point and does not substitute an arbitrary completed power series at a unit. The map Φ_0 itself is continuous on $\mathcal{R}$, because $u^p - a$ belongs to its maximal ideal; its derivative is the stated chain-rule expression.

I.13.3 Cross-multiply before approaching supersingularity

On the ordinary part of the3-isogeny correspondence, the canonical p-subgroups commute with phi3, since their defining connected kernels are preserved by prime-to-p isogenies. Comparing the normalized weight-two ratios therefore gives the exact identity

$$\rho(u)\,\sigma_C(M) = M\,\rho(v).$$

All factors can be checked intrinsically from $M = 9\Theta_s/\phi_3^*\Theta_t$: the factor nine and the p^2 in each normalized ratio cancel. Lemma I.11.1 and (I.13.6) therefore give

$$\Phi_0(M)B(u) + pu^pT(u)\Phi_0(M') - M(u)B(v(u)) \equiv 0 \pmod{p^2} \tag{I.13.7}$$

on the ordinary localization. Each term of (I.13.7) is regular in R, even at its supersingular center.

The congruence extends to R by p^2-saturation. Explicitly, the reduction of B(u) is a nonzero power series (it has exact order2), so it is a nonzero divisor in R/pR. If multiplication by a power of B makes a regular expression divisible by p^2, first reduce modulo p and cancel B, then divide by p and cancel B again modulo p. Thus R/p^2R injects into its ordinary localization, and (I.13.7) holds in R. This is the extension of a regular cross-multiplied expression; sigma itself is never specialized at the supersingular point.

I.13.4 Differentiate the regular identity

Differentiate (I.13.7) with respect to u and set u=a. Lemma I.9.1 gives B(a)=0 modp^2, B'(a)=0 modp, T(a)=0 modp. Terms involving B(a), or an additional pT(a), consequently disappear modulo p^2. Since Phi0 reduces to Frobenius and the values in (I.13.4) are rational,

$$\Phi_0(M)(a) = 3 \pmod{p}, \qquad \Phi_0(M')(a) = -54 \pmod{p}.$$

The derivative identity becomes precisely

$$3B'(a) - 54paT'(a) = -3B'(a) \pmod{p^2}.$$

As a=1/36 and $p > 5$, this simplifies to

$$\boxed{4B'(a) = pT'(a) \pmod{p^2}.} \tag{I.13.8}$$

Finally compare the double-pole coefficients at $u = a$ in (I.7.2). Write, over $\mathbb{F}_p$, $B = b_2(u-a)^2 + \cdots$, $T = t_1(u-a) + \cdots$, with $b_2 \neq 0$ by Lemma I.9.1. The coefficient of $(u-a)^{-2}$ gives

$$-a\frac{t_1}{b_2} = \frac{\Delta(a)^{(p-1)/2}}{b_2}, \qquad aT'(a) = -\Delta(a)^{(p-1)/2} \pmod{p}. \tag{I.13.9}$$

Since Delta(a)=50/81, its Legendre symbol is(2/p). Equations (I.13.8),(I.13.9) and B(a)=0 modp^2 yield

$$(1+4\theta)B(a) = 4aB'(a) = paT'(a) = -p(2/p) = p(-15/p) \pmod{p^2},$$

where the last equality uses the inert hypothesis $(-30/p) = -1$. This proves (I.13.1).

□

I.14 The weighted boundary cases p=3 and p=5

Lemma I.14.1 (SMALL)**.** $W_3(36) \equiv 0 \pmod 9$ *and* $W_5(36) \equiv 0 \pmod{25}$*, so the weighted congruence holds at* $p = 3, 5$*.*

Proof. Direct expansion of the defining fourth-power polynomials at36 gives

$$S_0(36) = 1, \quad S_1(36) = 37, \quad S_2(36) = 1873, \quad S_3(36) = 154549, \quad S_4(36) = 15312385.$$

For example, $S_2(36) = 1 + 16 \cdot 36 + 36^2$; the analogous fourth-power binomial expansions give the other displayed values. Consequently

$$W_3(36) = 7 + 15 \cdot 37 + 23 \cdot 1873 = 43641 = 9 \cdot 4849,$$
$$W_5(36) = W_3(36) + 31 \cdot 154549 + 39 \cdot 15312385 = 602017675 = 25 \cdot 24080707.$$

For each p in{3,5}, the Jacobi symbol (p/15)=0 under the Jacobi-symbol convention fixed above. Thus the right side of target II is zero and these two integer equalities prove the required congruence. They exhaust the small-prime boundary because every other odd prime exceeds5. Arithmetic was checked directly by exact integer expansion.

□

I.15 Completion of the congruence proofs

Proof of Theorems 1.1 and 1.2. For every $p > 5$, $(-30/p)$ is either 1 or -1. Lemma CM-POINT proves the zero branch of (1.1) in the latter case. Lemma I.12.1 proves all four representation branches in the former, including their existence and their exact character labels. This proves (1.1).

For (1.2), first take $p \geq 7$, and abbreviate

$$K = (-30/p), \qquad A = (2/p), \qquad J = (10/p), \qquad h = T/B.$$

Lemmas SPLIT and IRR, in their disjoint cases, give the single all-primes value

$$R(a) \equiv p(-15/p) \pmod{p^2}. \tag{I.15.1}$$

In particular $R(a) = 0 \pmod p$. If $K = 1$, Lemma I.10.1 makes $B(a)$ a unit; Lemma I.7.1 gives

$$(1 + 4\theta)T = hR + 4B\theta h \equiv 4A - 4B(a) \pmod p$$

at a, since $\Delta(a) = 50/81$ has Legendre symbol A. If $K = -1$, Lemma I.9.1 gives an exact double zero of B and $T(a) = 0$. Lemma I.6.1 and Lemma I.7.1 make T/B have at most a simple pole. Write, in $\mathbb{F}_p$ with $z = x - a$,

$$B = b_2 z^2 + O(z^3), \qquad T = t_1 z + O(z^2), \qquad b_2 \neq 0.$$

The coefficients of z^{-2} in Lemma I.7.1's rational differential identity give $-at_1 = A$. Thus in this case $(1+4\theta)T(a) = -4A \pmod p$, while $B(a) = 0 \pmod p$. The two cases can therefore be combined as

$$(1 + 4\theta)T(a) + 4B(a) = 4KA \pmod p. \tag{I.15.2}$$

Lemma I.8.1 gives $C(a) = J \pmod p$ and

$$W_p(36) \equiv 36^{p-1}\{-R(a) + p(1+4\theta)T(a) + 4pB(a) + 4pC(a)\} \pmod{p^2}.$$

By (I.15.1) the braces are divisible by p, so the factor 36^{p-1} can be replaced by one. Equations (I.15.1)–(I.15.2) give

$$W_p(36)/p \equiv -(-15/p) + 4KA + 4J = 3KA + 4J \pmod p,$$

because $(-15/p) = KA$. Quadratic reciprocity gives exactly

$$KA = \left(\frac{p}{15}\right), \qquad J = \left(\frac{p}{15}\right)\left(\frac{-6}{p}\right).$$

This is (1.2) for every $p \geq 7$; Lemma I.14.1 proves its two remaining odd-prime cases. Thus both assertions have their full stated scope.

□

I.16 Certificate appendix and reproducibility

Only exact symbolic identities or finite Fourier certificates accompanied by the stated valence reduction are proof premises. Finite prime tests and numerical approximations are not used. The following files are the precise essential computational certificates or optional symbolic expansion checks. The named files are supplied in the ancillary directory.

- PFLOG symbolic certificate source and its exact log check the displayed all-parameter polynomial identity. The displayed identity is also directly verifiable by polynomial expansion without running a program.
- TATE weight-twelve certificate, source, and log give every Fourier coefficient used through the inclusive valence bound 18.
- TATE weight-eight bridge certificate, source, and log give every coefficient used through the inclusive valence bound 12.
- POLES exact rational-model certificate and source check the resultant, deck transformations, discriminants, cusp orders, and rational derivatives displayed in the proof. These identities can equally be checked from the displayed formulas by exact algebra.
- CMPOINT exact polynomial certificate, source, and log check the diagonal factorization and exact germ derivatives in IRR. The modular equation itself is the established Cooper equation cited in CMPOINT, rather than an experimentally inferred relation.

The two essential finite valence checks are reproducible using Python 3.13.6 and exact integer/Fraction arithmetic, with a 30-second timeout per source. The symbolic checks use SymPy 1.14.0. For an independent check without those libraries, the product expansion rule for each integer exponent is

$$(1-Q^m)^r = \sum_{j\geq 0}(-1)^j\binom{r}{j}Q^{mj} \quad (r \in \mathbb{Z}),$$

and $E_2(Q) = 1 - 24\sum_{n\geq 1}\sigma_1(n)Q^n$. Truncating these products at the relevant bound gives the supplied finite arrays by integer arithmetic. The certificate hashes below bind the supplied arrays and named source versions, while the valence arguments establish why precisely those coefficients suffice for the general identities.

Part II
Divisibility for every positive integer

The statement proved is

$$W_n := \sum_{k=0}^{n-1}(8k+7)\sum_{j=0}^{k}\binom{k}{j}^4 36^j \quad \Longrightarrow \quad \frac{W_n}{n} \in \mathbb{Z} \qquad (n \geq 1). \tag{II.0.1}$$

All constant terms below are taken in the three displayed Laurent variables. The prime-adic arguments will prove $W_n/n \in \mathbb{Z}_p$ for **every** prime p and **every** positive integer n; neither the exponent of p nor the cofactor of n is bounded.

II.1 An exact Newton conversion

Put

$$\Lambda_s = (1+X)(1+Y)(1+Z)(1+s/(XYZ)), \qquad P = \Lambda_{36} - 1,$$
$$A_j = \mathrm{CT}(P^j) \quad (j \geq 0), \qquad A_{-1} = 0, \qquad D(h) = A_{h-1} + 8A_{h-2} \quad (h \geq 1). \tag{II.1.1}$$

Expanding the four factors of Λ_s^k shows that

$$\mathrm{CT}(\Lambda_s^k) = \sum_{j=0}^{k}\binom{k}{j}^4 s^j = S_k(s) : \tag{II.1.2}$$

to obtain exponent zero, the selected exponent in each of the first three factors must equal the selected exponent in the fourth. Since $\Lambda_{36} = 1 + P$, this gives

$$S_k(36) = \sum_{j=0}^{k}\binom{k}{j}A_j.$$

Use $k\binom{k}{j} = j\binom{k}{j} + (j+1)\binom{k}{j+1}$ and the two hockey-stick sums to obtain

$$W_n = \sum_{j=0}^{n-1}\left((8j+7)\binom{n}{j+1} + 8(j+1)\binom{n}{j+2}\right)A_j.$$

Dividing by n and reindexing its second term proves the exact rational identity

$$\frac{W_n}{n} = \sum_{j=0}^{n-1}\binom{n-1}{j}q_j, \qquad q_j = \frac{(8j+7)A_j + 8jA_{j-1}}{j+1}. \tag{II.1.3}$$

For $h = j + 1$, the numerator here is

$$8h(A_{h-1} + A_{h-2}) - D(h). \tag{II.1.4}$$

Consequently, for a fixed prime p, the conditions

$$D(h)/h \in \mathbb{Z}_p \qquad (h \geq 1) \tag{II.1.5}$$

imply $q_j \in \mathbb{Z}_p$ for every j, and hence $W_n/n \in \mathbb{Z}_p$ for every n. The boundary values are $D(1) = 1$ and $q_0 = 7$. We will use (II.1.5) at primes $p \geq 5$. The primes 2 and 3 have a simpler direct proof.

II.2 The prime divisors of 36

More generally, if a prime p divides an integer s, then

$$p^{v_p(n)} \mid \sum_{k<n} (8k+7) S_k(s) \qquad (n \geq 1). \tag{II.2.1}$$

Here and below v_p is the usual valuation, with $v_p(0) = +\infty$. To prove this, extend $\binom{k}{j}$ by zero for $j > k$. If $p^r \mid h$, Vandermonde and $i\binom{h}{i} = h\binom{h-1}{i-1}$ give, for $j \geq 1$,

$$v_p\left(\binom{k+h}{j} - \binom{k}{j}\right) \geq r - \lfloor \log_p j \rfloor.$$

Multiplying the corresponding difference of fourth powers by s^j raises this lower bound to $r + j - \lfloor \log_p j \rfloor \geq r + 1$. The $j = 0$ terms cancel, so

$$S_{k+h}(s) \equiv S_k(s) \pmod{p^{r+1}}. \tag{II.2.2}$$

Induct on $a = v_p(n)$, uniformly in the cofactor. Write $n = pm$, where $v_p(m) = a - 1$, and split the sum into the p blocks $k + \ell m$, $0 \leq k < m$, $0 \leq \ell < p$. Equation (II.2.2) gives

$$W_{pm}(s) \equiv pW_m(s) + 4mp(p-1) \sum_{k<m} S_k(s) \pmod{p^a}.$$

The last term is divisible by p^a, also when $p = 2$, and the first is divisible by p^a by induction. This proves (II.2.1), and treats 2 and 3 at $s = 36$.

II.3 The geometric family and its integral models

The geometric calculations needed later are supplied here, including those at the singular prime 5. For units s, t over a base ring R, let $X_{s,t}$ be the complete intersection in $V = (\mathbb{P}^1_R)^4$

$$F_1 = \prod_{i=1}^{4} X_i - s \prod_{i=1}^{4} Y_i = 0, \qquad F_2 = \prod_{i=1}^{4} (X_i + Y_i) - t \prod_{i=1}^{4} Y_i = 0. \tag{II.3.1}$$

On its torus, with $x_i = X_i/Y_i$, these are $\prod x_i = s$, $\prod(1 + x_i) = t$. Eliminating $x_4 = s/(XYZ)$ gives $\Lambda_s - t = 0$.

The first equation defines the integral split toric threefold W_s, which becomes independent of s after scaling the fourth coordinate. The second section is nonzero on every geometric fiber of W_s: at its toric points with two coordinates zero and two infinite, its value in the natural homogeneous trivialization is 1. It is thus a fiberwise nonzerodivisor. Both equations are relative effective Cartier equations; their complete intersection is flat over the unit-parameter base. Write $H = H_1 + H_2 + H_3 + H_4$. The Koszul resolution has terms $\mathcal{O}_V$, $\mathcal{O}_V(-H)^2$, and $\mathcal{O}_V(-2H)$. Künneth and adjunction give, universally under base change,

$$H^0(\mathcal{O}_X) = R, \quad H^1(\mathcal{O}_X) = 0, \quad H^2(\mathcal{O}_X) = R, \quad \omega_X \simeq \mathcal{O}_X. \tag{II.3.2}$$

These calculations do not require smoothness. Every smooth geometric fiber is therefore a K3 surface.

II.3.1 Boundary and discriminant

Every point off the torus has one coordinate ∞, a different coordinate 0, and a third coordinate -1. These conditions give 24 boundary lines, indexed by the assignments of the four roles

$$(i_\infty, j_0, k_{-1}, \ell_{\text{free}}) \tag{II.3.3}$$

to the four indices. A local calculation covers even their intersections. Take $x_1 = \infty, x_2 = 0, x_3 = -1$; put $z = 1/x_1, v = x_2, w = x_3 + 1, x_4 = a$. The equations are

$$v(-1+w)x_4 - sz = 0, \qquad (1+z)(1+v)w(1+x_4) - tz = 0.$$

Along $z = v = w = 0$, the minors in columns (z, w) and (v, z) are $-s(1+a)$ and at. They cover all finite a, including $a = -1$. At $a = \infty$, put $d = 1/x_4$; the equations become

$$v(-1+w) - szd = 0, \qquad (1+z)(1+v)w(1+d) - tzd = 0,$$

with minor -1 in columns (v, w). The boundary is smooth in every characteristic with unit s, t. Two of its lines meet transversely exactly when their role labels differ by swapping the free role with one of the other three roles. The intersection parameters of a line are its free coordinate equal to $\infty, 0, -1$; these are all its intersections.

On the torus the two Jacobian rows are s/x_i and $t/(1+x_i)$. Their dependence forces all four $x_i = q$, and the critical parameters are exactly

$$s = q^4, \qquad t = (1+q)^4. \tag{II.3.4}$$

For $s = 1$, their eliminant is

$$\operatorname{Res}_q(q^4 - 1, (1+q)^4 - t) = t(t-16)(t+4)^2. \tag{II.3.5}$$

At $t = 36$, this equals $2^{10}3^2 5^3$. Thus $X_{1,36}$, and its exchanged model $X_{36,1}$, are smooth proper K3 surfaces over $\mathbb{Z}[1/30]$. The same holds in open parameter neighborhoods of these sections.

II.3.2 The toric resolution and residue normalization

The toric threefold W_s has cocharacter lattice $N = \{(a_1, a_2, a_3, a_4) \in \mathbb{Z}^4 : \sum a_i = 0\}$, with rays $e_i - e_j$ for $i \neq j$. Its fan is the intersection of the orthant fan with $N_{\mathbb{R}}$. Eight maximal cones, of sign types $1+3$ and $3+1$, are smooth. The other six, of type $2+2$, have the four rays

$$e_i - e_j,\ e_i - e_\ell,\ e_k - e_j,\ e_k - e_\ell.$$

Their only relation is the equality of the two opposite sums. They are ordinary-node cones; equivalently W_s has precisely six nodes, with local equation $ab = scd$, where two coordinates are zero and two infinite.

Triangulate each square using its existing rays. The triangulations may be chosen coherently: perturb a strictly convex rational support function by sufficiently small multiples of ij at the ray $e_i - e_j$. The opposite-sum difference on a square is $(i-k)(j-\ell) \neq 0$. For a sufficiently small perturbation the old walls retain convexity, and all six squares acquire regular triangulations. Every resulting three-ray cone is unimodular: its rays give the incidence matrix of a spanning tree on four vertices, hence a basis of N. We obtain a smooth projective toric small crepant resolution $\widetilde{W}_s \to W_s$ over the unit-parameter base; there are no new rays or exceptional divisors.

The section F_2 has value 1 at all six nodes, so X avoids them. Its strict transform is the same surface. Adjunction gives $-K_{W_s} = \mathcal{O}(H)|_{W_s}$; consequently the surface is anticanonical

in $\widetilde{W}_s$. The coefficients of its toric divisor are all 1. A fixed point above any node is avoided by the surface and has unit section value; this supplies a unit vertex coefficient. Equivalently one may use a smooth $1+3$ cone, whose vertex coefficient is 1 or s.

The Newton polytope of $\Lambda_s - t$ is

$$\Delta = \mathrm{conv}([0,1]^3 \cup [-1,0]^3). \tag{II.3.6}$$

It contains the cross-polytope with vertices $\pm e_i$, so 0 is interior. Containment in $[-1,1]^3$ excludes every other interior integral point. The polarized normal toric variety of Δ is exactly W_s: the characters of the restrictions of the 16 degree-$(1,1,1,1)$ monomials are the images of the four-dimensional unit cube in $\mathbb{Z}^4/\mathbb{Z}(1,1,1,1)$, which give (II.3.6), and the restricted polarization is ample.

In particular the constant-term matrices for the unique interior point are scalars

$$\beta_m(s,t) = \mathrm{CT}(\Lambda_s - t)^{m-1}. \tag{II.3.7}$$

The residue basis has no missing parameter factor:

$$\omega_{s,t} = \mathrm{Res}_{X_{s,t}} \frac{d\log X \wedge d\log Y \wedge d\log Z}{\Lambda_s - t}. \tag{II.3.8}$$

Eliminating the fourth variable in the double residue of $dx_1 \wedge \cdots \wedge dx_4/(F_1F_2)$ gives exactly (II.3.8), up to a fixed orientation sign.

The two global projective coordinate transformations

$$E : x_i \mapsto -1 - x_i, \qquad I : x_i \mapsto 1/x_i$$

induce base maps

$$E(s,t) = (t,s), \qquad I(s,t) = (1/s, t/s). \tag{II.3.9}$$

Their residue factors, also valid for the dualizing form of singular complete intersections, are

$$E^*\omega_{t,s} = -\omega_{s,t}, \qquad I^*\omega_{1/s,t/s} = -s\omega_{s,t}. \tag{II.3.10}$$

Indeed exchange swaps the two equations. Inversion pulls them back to $-F_1/s$ and $F_2 - (t/s)F_1$ in homogeneous trivializations. The product of the four projective coordinate Jacobians is 1; dividing by the determinant of the equation change gives -1 and $-s$.

II.3.3 The Hodge local-freeness required for the ordinary comparison

Here $p > 5$ and the family is restricted to its smooth locus. The boundary lines have square -2 by adjunction and the intersection rule of §3.1. The toric divisor D_{ab}, $a \neq b$, restricts to the sum of the two lines with infinity at a, zero at b, and the remaining roles in either order. The three primitive character relations are $\sum_b D_{ab} - \sum_b D_{ba} = 0$, with three independent choices of a. Removing D_{14}, D_{24}, D_{34} gives an integral basis of the rank-nine toric Picard group. In the order

$$12, 13, 21, 23, 31, 32, 41, 42, 43$$

the intersection matrix of their restrictions is

$$Q=\begin{pmatrix}
-2 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0\\
1 & -2 & 0 & 1 & 0 & 0 & 0 & 0 & 1\\
0 & 0 & -2 & 1 & 1 & 0 & 1 & 0 & 0\\
0 & 1 & 1 & -2 & 0 & 0 & 0 & 0 & 1\\
0 & 0 & 1 & 0 & -2 & 1 & 1 & 0 & 0\\
1 & 0 & 0 & 0 & 1 & -2 & 0 & 1 & 0\\
0 & 0 & 1 & 0 & 1 & 0 & -2 & 1 & 1\\
1 & 0 & 0 & 0 & 0 & 1 & 1 & -2 & 1\\
0 & 1 & 0 & 1 & 0 & 0 & 1 & 1 & -2
\end{pmatrix}, \qquad \det Q = 128. \tag{II.3.11}$$

Its Smith factors are $1,1,1,1,1,1,4,4,8$. The determinant alone shows that the toric restriction is split injective at every odd prime: invert its pairing matrix, using the perfect Serre pairing on $H^1(X,\Omega^1_X)$. Its dual Gysin map is therefore split surjective.

For completeness, the remaining freeness hypotheses may be checked directly. Koszul gives (II.3.2) and $H^i(\mathcal{O}_X(-H))=0$ for $i=0,1$. The conormal sequence

$$0 \longrightarrow \mathcal{O}_X(-H)^2 \longrightarrow \Omega^1_V|_X \longrightarrow \Omega^1_X \longrightarrow 0$$

gives $H^0(\Omega^1_X)=0$: the middle summands $\mathcal{O}_X(-2H_i)$ have no sections, again by Koszul. Serre duality gives $H^2(\Omega^1_X)=0$. The Chern calculation gives $\deg c_2(X)=24$, and surface Riemann–Roch yields $\chi(\Omega^1_X)=-20$. Thus $H^1(\Omega^1_X)$ has constant dimension 20. Together with (II.3.2) and duality, this proves local freeness and base change for the Hodge groups of X.

For the smooth projective toric threefold $\widetilde{W}$, $\mathcal{O}$ and the structure sheaf of each toric divisor have no positive-degree cohomology, universally over the split base. The sequence for $\mathcal{O}(-D_\rho)$, the toric Euler sequence and Serre duality then give diagonal free Hodge groups of ranks $1,9,9,1$, and zero off the diagonal. Finally

$$(-K_{\widetilde{W}})^3 = H^4 \text{ on } (\mathbb{P}^1)^4 = 24.$$

The class of X is primitive at $p>3$, since its cube is a unit there. In the residue sequences

$$0 \longrightarrow \Omega^i_{\widetilde{W}} \longrightarrow \Omega^i_{\widetilde{W}}(\log X) \longrightarrow \Omega^{i-1}_X \longrightarrow 0, \tag{II.3.12}$$

the potentially nonzero maps are the primitive map sending 1 to $[X]$, the Gysin map dual to the split toric restriction in (II.3.11), and the top trace isomorphism. Their kernels and cokernels are free. This verifies the Hodge local-freeness and base-change hypotheses for the logarithmic pair as well. These arguments are valid near both required smooth sections at every $p>5$.

II.4 Every ordinary prime, to every order

Fix $p>5$ with $\left(\frac{-30}{p}\right)=1$. We use the following two established comparison results.

- Beukers–Vlasenko, *Dwork crystals I*, arXiv:1903.11155v4, Theorems 4.3 and 5.3: on a p-complete, p-separated characteristic-zero coefficient domain, a Laurent polynomial with a unit vertex and invertible Hasse matrix has a quotient by formal derivatives of the expected interior-point rank. If that rank is one, its connection scalar N_δ, in the basis $1/f$, satisfies

$$\delta\beta_{mp^r} \equiv N_\delta \beta_{mp^r} \pmod{p^r} \quad (m, r \geq 1), \qquad N_\delta = \lim_{r\to\infty} \frac{\delta\beta_{p^r}}{\beta_{p^r}}. \tag{II.4.1}$$

The derivation acts on the coefficient ring before specialization. The quantifier over m is part of the theorem; β_{mp^r} is not assumed to be a unit.

- Huang–Lian–Yau–Yu, *Hasse–Witt matrices, unit roots and period integrals.* arXiv:1801.01189, §5 and Theorem 5.3: for a smooth hypersurface in a smooth complete toric ambient space, with positive toric divisor coefficients, the Hodge local-freeness conditions for hypersurface and logarithmic pair, invertible Hasse–Witt matrix, and an ambient toric fixed point avoided by the hypersurface, the same limiting connection is the intrinsic crystalline **top-slope quotient** connection in the residue basis. In particular it is natural under isomorphisms of the proper smooth hypersurfaces, with the scalar change dictated by their residue forms. The proof of Theorem 5.3 establishes the comparison; its intermediate Conjecture 5.4 is not an extra premise.

Apply these results on the p-adic completion of the smooth two-parameter base with β_p inverted. This is a coefficient domain. A Frobenius lift on the polynomial parameters extends to the completed localization, since it takes each inverted element to a unit modulo p. The derivations ∂_s, ∂_t extend continuously. Sections 3.2–3.3 checked the unit vertex, smooth projective toric realization, positive anticanonical coefficients and both Hodge hypotheses. There is exactly one interior point; write the two scalar connection coefficients as N_s, N_t.

For the ordinary condition, put

$$f_k = \sum_{j=0}^{k} \binom{k}{j}^4, \qquad G(x) = \sum_{k\geq 0} f_k x^k, \qquad B_p(x) = \sum_{k=0}^{p-1} f_k x^k, \qquad a = 1/36. \tag{II.4.2}$$

Binomial expansion gives $\beta_p(1, 36) \equiv 36^{p-1} B_p(a) \pmod{p}$, and $\beta_p(36, 1) \equiv U_p \equiv B_p(a) \pmod{p}$, the last equality being Lemma I.2.1. The Hasse criterion (I.10.10) states that $B_p(a) = 0 \pmod{p}$ precisely when $\left(\frac{-30}{p}\right) = -1$. Thus both values are units here.

The intrinsic basis in the comparison is the residue image in the top-slope quotient. No invariance of the holomorphic line under the full Gauss–Manin connection is asserted. The scalar convention is exactly the positive beta-derivative limit in (II.4.1).

II.4.1 The two geometric symmetries

Functoriality and (II.3.10) give

$$N_s(s, t) = N_t(t, s), \tag{II.4.3}$$

and, after applying inversion and differentiating its scalar factor,

$$N_s(s, t) = -\frac{N_s(1/s, t/s)}{s^2} - \frac{tN_t(1/s, t/s)}{s^2} - \frac{1}{s}.$$

At $s = 1$, therefore,

$$2N_s(1, t) = -1 - tN_t(1, t). \tag{II.4.4}$$

The term -1 is the logarithmic derivative of the residue factor; it is essential.

II.4.2 Cusp normalization

On $s = 1$, put $x = 1/t$ and initially, where $x \neq 0$, define

$$g_p(x) = -t^2 N_t(1, t) - t. \tag{II.4.5}$$

This function continues regularly to the ordinary x-line at zero, with cusp expansion G'/G. To check the cusp without assuming a unit vertex after a coefficient vanishes, set

$$H_m(x) = \mathrm{CT}(1 - x\Lambda_1)^{m-1} = \sum_{k<m} \binom{m-1}{k} (-x)^k f_k.$$

For $m = p^b$, direct differentiation gives

$$-t^2 \frac{\partial_t \beta_m(1, t)}{\beta_m(1, t)} - t = \frac{H'_m}{H_m} - \frac{p^b}{x}. \tag{II.4.6}$$

Modulo p, H_{p^b} is a product of Frobenius transforms of H_p, and $H_p \equiv B_p$. Thus the ratios H'_{p^b}/H_{p^b} lie in the p-adic completion of $\mathbb{Z}_p[x, B_p^{-1}]$, and are regular at zero. By (II.4.1) and (II.4.6) they are compatible modulo p^b where $x \neq 0$. Compatibility extends across zero: x is a nonzerodivisor modulo p, so localization at x is p^r-saturated, by canceling x modulo p and then canceling p successively. This gives the continuation. One may also invert the smooth discriminant, a unit at zero.

For fixed k, $(-1)^k \binom{p^b-1}{k} \to 1$ in $\mathbb{Z}_p$. The constant terms of all H_{p^b} are 1, so coefficientwise inversion and differentiation commute with this limit. Hence

$$g_p(x) = G'(x)/G(x) \quad \text{as a cusp power series.} \tag{II.4.7}$$

Modulo every p^r, g_p is a rational function with invertible denominator on the ordinary smooth locus. We have continued a connection, without evaluating a period series at a p-adic unit.

II.4.3 The exact CM value

Lemmas I.4.1 and I.13.2 give the normalized modular correspondence

$$u = x(\tau), \quad v = x(3\tau), \quad M = \Theta(\tau)/\Theta(3\tau), \qquad G(u) = MG(v) \tag{II.4.8}$$

at its selected cusp, where $v = u^3 + \cdots$, and its rational normalized CM point with

$$u = v = a, \qquad M = 3, \quad M'_u = -54, \quad v'_u = -1. \tag{II.4.9}$$

The normalized curve is the tame quotient of $X_0(30)$ by the prime-to-3 Atkin–Lehner involutions. It is p-flat with geometrically integral reduction for $p > 5$. Its multiplier is the algebraic function with intrinsic normalization $9\Theta(E)/\varphi_3^*\Theta(E/C_3)$, giving exactly (II.4.8). These integral objects are part of IRR's construction. At the CM point both partial derivatives of the plane equation are $4375/236196$. This is a unit at every current ordinary prime: its only prime factor exceeding 5 is 7, and 7 is inert. The branch is therefore an integral étale source-u chart. Both Theta sections are units there by the Hasse identity in POLES and the ordinary criterion. The degree-three isogeny differential is a unit, so M, M^{-1}, M'_u, v'_u are regular there.

Differentiate the cusp identity (II.4.8) and use (II.4.7):

$$g_p(u) = M'_u/M + v'_u g_p(v). \tag{II.4.10}$$

The identity extends modulo every p^r. Indeed its terms modulo p^r are rational functions on the correspondence, regular on a suitable open locus containing the cusp and this CM point. Cross-multiply their finitely many denominators. Modulo p, a zero cusp expansion forces the rational difference to vanish, since the reduced curve is integral. The difference is divisible by p; divide by p using flatness and repeat. This proves the equality modulo p^r for all r, hence exactly after evaluation in $\mathbb{Z}_p$.

At (II.4.9) the identity reads

$$g_p(a) = -18 - g_p(a), \qquad g_p(a) = -9 \text{ in } \mathbb{Z}_p. \tag{II.4.11}$$

Equations (II.4.3)–(II.4.5) now give

$$N_t(1,36) = -\frac{1}{48}, \qquad N_s(1,36) = -\frac{1}{8}, \qquad N_t(36,1) = -\frac{1}{8}. \tag{II.4.12}$$

II.4.4 Every cofactor and exponent

Take any $h = mp^r$ with $r = v_p(h) \geq 1$. Equation (II.4.1), for $\delta = \partial_t$ at $(36,1)$, and (II.4.12) give $\beta_h + 8\partial_t\beta_h \equiv 0 \pmod{p^r}$. Since $\beta_h = A_{h-1}$ and $\partial_t\beta_h = -(h-1)A_{h-2}$, it follows that

$$D(h) = A_{h-1} + 8A_{h-2} \equiv 0 \pmod{p^r}. \tag{II.4.13}$$

For h prime to p, (II.1.5) is automatic. Therefore (II.1.5), and thus $W_n/n \in \mathbb{Z}_p$, hold for every ordinary $p > 5$ and every n.

II.5 The constant-term formal group, including nonreduced bases

The inert primes and 5 will use an integral formal group and its first parameter derivative. We give the comparison with the actual formal Brauer functor, since identifying only its tangent space or only its reduced fiber would not suffice.

Dingxin Zhang, *On Vlasenko's formal group laws*, arXiv:2009.10512, Theorem 1.4 and Corollary 1.17, supplies the following theorem. A full-dimensional Laurent polynomial over a flat $\mathbb{Z}$-algebra defines an integral formal group with its specified constant-term logarithm; its dimension is the number of interior lattice points. Over a noetherian base its functor is the degree-dimension Artin–Mazur group of the hypersurface **ideal sheaf** on the associated projective toric scheme. Neither smoothness of the hypersurface nor reduction of the base is a hypothesis. Remark 0.2 distinguishes this functor from the formal Brauer group of the hypersurface in general.

Here (II.3.6) has one interior point, and the coefficient logarithm is

$$\ell_{s,t}(T) = \sum_{h\geq 1} \beta_h(s,t)\frac{T^h}{h}, \qquad \beta_h(s,t) = \mathrm{CT}(\Lambda_s - t)^{h-1}. \tag{II.5.1}$$

We will use the theorem over $R = \mathbb{Z}_p[\epsilon]/\epsilon^2$, and at 5 over $R = \mathbb{Z}_5[[S, T_0]]$. These are noetherian and $\mathbb{Z}$-flat. The symbol T in (II.5.1) is the formal-group coordinate, independent of the base parameters. Section 3 identifies its toric ambient with W_s and its hypersurface with the flat complete intersection (II.3.1).

Here is the required additional identification. Set $I = \mathcal{O}_{W_s}(-X_{s,t})$. There is the exact sequence

$$0 \longrightarrow \mathcal{O}_{(\mathbb{P}^1)^4}(-H) \longrightarrow \mathcal{O}_{(\mathbb{P}^1)^4} \longrightarrow \mathcal{O}_{W_s} \longrightarrow 0.$$

The first sheaf has no cohomology in any degree; the second has only $H^0 = R$, universally after tensoring with any coefficient module. Thus

$$H^q(W_s, \mathcal{O}_{W_s} \otimes_R M) = 0 \qquad (q > 0) \tag{II.5.2}$$

for every R-module M. This direct Koszul calculation applies to the original singular toric ambient, so no resolution or smoothness assumption is hidden in it.

For a nilpotent R-algebra J, flatness of X gives an exact sequence of abelian sheaves

$$1 \longrightarrow 1 + I \otimes_R J \longrightarrow 1 + \mathcal{O}_{W_s} \otimes_R J \longrightarrow 1 + \mathcal{O}_X \otimes_R J \longrightarrow 1. \tag{II.5.3}$$

The last map is locally surjective by lifting nilpotent functions. Filter its middle sheaf by powers of J; the successive quotients are the quasicoherent sheaves $\mathcal{O}_{W_s} \otimes_R (J^b/J^{b+1})$. Equation (II.5.2) makes its positive-degree cohomology vanish. The connecting map is consequently a natural isomorphism

$$H^2(X, 1 + \mathcal{O}_X \otimes_R J) \simeq H^3(W_s, 1 + I \otimes_R J). \tag{II.5.4}$$

Nilpotent-unit cohomology is the same in the Zariski and étale topologies, by the quasicoherent filtration. Its H^2 is the infinitesimal formal Brauer functor: for a split nilpotent coefficient extension every line bundle on the unthickened scheme pulls back to the thickening, so the Picard connecting obstruction in the usual kernel definition vanishes. Thus (II.5.4) identifies Zhang's group with the actual formal Brauer functor, including on these nonreduced bases and for singular flat fibers. It is an isomorphism of functors, natural under family isomorphisms. Arbitrary nil algebras may be treated by filtered colimits; all arguments about coefficients of a power series need only finite nilpotent T-truncations.

We will repeatedly use the following elementary consequence.

Lemma II.5.1 (Integral logarithm)**.** *Let A be p-torsion-free and p-local, and let F be a one-dimensional integral formal group over A. If its entire p-series belongs to $pA[[T]]$, then its normalized characteristic-zero logarithm belongs to $A[[T]]$.*

Proof. Write $[p]_F(T) = Q(T)$, $Q_1 = p$, and $\ell(T) = T + \sum_{h\geq 2} \ell_h T^h$. The identity $\ell(Q(T)) = p\ell(T)$ gives

$$(p - p^h)\ell_h = Q_h + \sum_{2\leq j<h} \ell_j [T^h] Q(T)^j. \tag{II.5.5}$$

If the smaller ℓ_j are integral, the right side lies in pA. The factor $p - p^h = p(1 - p^{h-1})$ is p times a unit; cancel p in the torsion-free rational overring and then that unit. Induction proves the lemma, also for $A = \mathbb{Z}_p[\epsilon]/\epsilon^2$.

□

In particular, suppose the formal Brauer p-series of $X_{36,1+\epsilon}$ vanishes modulo p. By (II.5.1), (II.5.4) and the lemma, the coefficients of

$$\sum_{h\geq 1} \mathrm{CT}(P - \epsilon)^{h-1} \frac{T^h}{h}$$

belong to $\mathbb{Z}_p[\epsilon]/\epsilon^2$. Comparing their constant and ϵ parts gives, for every $h \geq 2$,

$$\frac{A_{h-1}}{h} \in \mathbb{Z}_p, \qquad \frac{(h-1)A_{h-2}}{h} \in \mathbb{Z}_p. \tag{II.5.6}$$

If $p \mid h$, then $h - 1$ is a unit, so both A_{h-1}/h and A_{h-2}/h are integral. If $p \nmid h$, they are integral automatically. Hence (II.5.6) implies (II.1.5), and by (II.1.3) proves $W_n/n \in \mathbb{Z}_p$ for

all n. What remains at each such prime is to prove vanishing of the **entire first-order** p-series, rather than just the central Hasse invariant.

II.6 The inert good fibers have Artin invariant one

Let $p > 5$ with $\left(\frac{-30}{p}\right) = -1$, and work over $k = \overline{\mathbb{F}}_p$. Section 3 makes $X = X_{1,36}$ a smooth proper K3 surface. Its Hasse invariant vanishes by (I.10.10) and the constant-term Hasse identification used in §4. Inversion $\iota : x_i \mapsto 1/x_i$ acts freely: a fixed ambient point has every $x_i = \pm 1$; these would give $t = 0$ if some sign is negative, or $t = 16$ otherwise. The actual value $t = 36$ is neither in characteristic $p > 5$.

The degree-two étale quotient is an Enriques surface. Its numerical lattice is $U \oplus E_8(-1)$, and its pullback is an invariant algebraic sublattice

$$L_+ \simeq U(2) \oplus E_8(-2), \qquad \operatorname{rank} L_+ = 10, \qquad |\det L_+| = 2^{10}. \tag{II.6.1}$$

These standard Enriques lattice facts, including odd characteristic, are stated in Dolgachev, *A brief introduction to Enriques surfaces*, §3. The quotient is actual and étale here; we are not using a characteristic-zero lattice embedding as a substitute for characteristic-p algebraic cycles.

Here is a nine-class anti-invariant lattice with an explicit small certificate. Label the boundary lines by permutations of the zero-based labels $0, 1, 2, 3$, in the role order $(\infty, 0, -1, \text{free})$. Inversion swaps the first two roles. For the nine permutations

$$0123,\ 0132,\ 0213,\ 0231,\ 0312,\ 1203,\ 1230,\ 1302,\ 2301, \tag{II.6.2}$$

take the actual divisor differences $d_c = C_c - C_{\iota c}$. Their intersection matrix is

$$R_- = \begin{pmatrix} -4 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 2 & -4 & 0 & 2 & 0 & 0 & -2 & 0 & 0 \\ 0 & 0 & -4 & 2 & 2 & 0 & 0 & 0 & 0 \\ 0 & 2 & 2 & -4 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 & -4 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -4 & 2 & 2 & -2 \\ 0 & -2 & 0 & 2 & 0 & 2 & -4 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & -4 & 2 \\ 0 & 0 & 0 & 0 & 0 & -2 & 0 & 2 & -4 \end{pmatrix}, \qquad \det R_- = -5120 = -2^{10} \cdot 5. \tag{II.6.3}$$

For example its successive LDL^t pivots are $-4, -3, -4, -5/3, -12/5, -4, -4/3, -9/4, -20/9$; their product verifies the determinant. Formula (II.6.3) follows directly from the boundary intersection rule in §3.1. Its nonzero determinant proves independence of these nine actual divisor classes, without making any assertion that a radical of a redundant Gram matrix is a divisor relation. Their lattice L_- is nondegenerate over $\mathbb{Z}_p$ for every $p > 5$. It is orthogonal to L_+, since ι fixes one and negates the other. Thus

$$L = L_+ \perp L_- \subset \mathrm{NS}(X) \tag{II.6.4}$$

is an actual rank-19 lattice with unit determinant over $\mathbb{Z}_p$. In particular $\rho(X) \geq 19$.

For a K3 surface of finite formal-Brauer height h, the bound $\rho + 2h \leq 22$ is Theorem 3.1 and Corollary 10.4 of van der Geer–Katsura, *On a stratification of the moduli of K3 surfaces*, arXiv:math/9910061v2. Hasse zero excludes $h = 1$. A finite height would therefore give $\rho \leq 18$, contradicting (II.6.4). Hence the height is infinite. Charles, *The Tate conjecture*

for K3 surfaces over finite fields, Theorem 1, implies $\rho = 22$ for such a supersingular K3 in characteristic at least 5. The surface here is defined over a finite field, as required.

The supersingular lattice theorem of Artin–Rudakov–Shafarevich says that in odd characteristic its Néron–Severi discriminant group is $(\mathbb{Z}/p)^{2\sigma}$, $1 \le \sigma \le 10$. A precise statement is recalled in Shimada, *Transcendental lattices and supersingular reduction lattices of a singular K3 surface*, Theorem-Definition 1.0.4. We use this general lattice theorem, not a separate CM reduction theorem from that paper.

Since (II.6.4) has unit Gram determinant, orthogonal projection with its inverse Gram matrix splits

$$\mathrm{NS}(X) \otimes \mathbb{Z}_p = (L \otimes \mathbb{Z}_p) \perp K, \qquad \operatorname{rank} K = 3.$$

Its p-primary discriminant group is that of K, and can be generated by at most three elements. Thus $2\sigma \le 3$, so $\sigma = 1$. Exchange identifies $X_{36,1}$ with X; it too has Artin invariant one.

II.7 A first-order Čech lemma and every inert primary factor

Lemma II.7.1 (First-order vanishing)**.** *Let X_0 be a smooth proper K3 surface over a perfect field k of characteristic $p > 2$, of height infinity, with*

$$\operatorname{Im}\bigl(H^1(X_0, B_r\Omega^1_{X_0}) \longrightarrow H^1(X_0, \Omega^1_{X_0})\bigr) = 0 \qquad (r \ge 1). \tag{II.7.1}$$

Here $B_1\Omega^1 = d\mathcal{O}$, and B_{r+1} is the inverse Cartier pullback of B_r, as usual. For a supersingular K3 of Artin invariant one, (II.7.1) *is exactly the consequence of Corollary 11.12 of van der Geer–Katsura, math/9910061v2: the image has dimension $\sigma - 1$ for every $r \ge \sigma$. Only this assertion about the central smooth fiber is used.*

Under (II.7.1)*, the entire p-series of the relative formal Brauer group of every smooth proper first-order deformation X_ϵ over $k[\epsilon]/\epsilon^2$ is zero.*

Proof. Fix $N \ge 2$, holding the formal-group variable T constant under spatial differentiation. Cover X_0 by affine opens with affine intersections. Every smooth affine first-order deformation is trivial, because its differential module is projective and $\operatorname{Ext}^1(\Omega^1, \mathcal{O}) = 0$. Choose trivializations of the lifted opens. Their transition maps are

$$\varphi_{ij} = 1 + \epsilon D_{ij},$$

where D_{ij} represents the Kodaira–Spencer class in $H^1(T_{X_0})$. The sheaf $U_N = 1 + T\mathcal{O}[T]/T^N$ has a finite filtration with quasicoherent additive quotients; our affine cover is acyclic for it. It computes the infinitesimal Brauer functor by the split-product observation used in §5.

Represent any formal Brauer value on X_ϵ by a multiplicative Čech 2-cocycle $a_\epsilon(T)$, and let a_0 be its reduction. In characteristic p,

$$a_\epsilon(T)^p = a_0(T)^p$$

chartwise: all ϵ-parts disappear. Central height infinity means that the central p-series is identically zero. Hence a central multiplicative 1-cochain $b_{ij}(T) \in U_N$ satisfies

$$\delta_0 b = a_0^p. \tag{II.7.2}$$

Lift its coefficients constantly in the chosen charts. With overlap sections transported to the first chart, the Čech differential is $(\delta_\epsilon b)_{ijk} = b_{ij}\varphi_{ij}(b_{jk})b_{ik}^{-1}$. It gives exactly

$$\frac{(\delta_\epsilon b)_{ijk}}{a^p_{\epsilon,ijk}} = 1 + \epsilon\, D_{ij}(b_{jk})/b_{jk} = 1 + \epsilon\langle D_{ij}, d\log b_{jk}\rangle. \tag{II.7.3}$$

The remaining obstruction to trivializing the p-th power is therefore the contraction cup product of the Kodaira–Spencer class with $[d\log b]$, coefficient by coefficient.

We check that each such coefficient is killed by (II.7.1). For a truncated unit, recursive removal of its coefficients gives the finite Euler factorization

$$b(T) = \prod_{1\le n<N} (1 - a_n T^n) \pmod{T^N}.$$

Spatial logarithmic differentiation gives

$$[T^m] d\log b = -\sum_{n|m} a_n^{m/n-1} da_n. \tag{II.7.4}$$

There is no factor n in this formula, because $dT = 0$. For a regular function a, Cartier sends $a^{pq-1}da$ to $a^{q-1}da$; if $p \nmid q$, the latter is the exact form $q^{-1}d(a^q)$ and is killed by Cartier. Thus $a^{q-1}da$ belongs to $B_{v_p(q)+1}\Omega^1$, with all intermediate Cartier images closed. Formula (II.7.4) belongs to $B_{v_p(m)+1}\Omega^1$.

Applying $d\log$ to (II.7.2) gives $\delta_0(d\log b) = d\log(a_0^p) = 0$. Its coefficient in degree m is consequently a 1-cocycle in that B-sheaf. The B_r-sheaves are coherent with the usual Frobenius-twisted module structures, so affine intersections compute their cohomology as well. By (II.7.1) its image in $H^1(\Omega^1)$ is zero. Its contraction with the Kodaira–Spencer class is therefore zero in $H^2(\mathcal{O})$. This holds for every $m < N$. The residual additive 2-cocycle in (II.7.3) is a coboundary; multiplying b by a suitable $1 + \epsilon h(T)$ makes its Čech differential exactly a_ϵ^p. Thus p times every value on $k[\epsilon, T]/(\epsilon^2, T^N)$ is zero.

Apply this to the universal formal-group coordinate and let N vary. Every finite truncation of the p-series vanishes, hence the entire series vanishes. No compatible choices of the cochains for different N are required. This proves the first-order vanishing lemma without an assertion about injectivity of relative Witt Frobenius over a nonreduced base.

□

At an inert $p > 5$, §6 gives Artin invariant one for $X_{36,1}$. The actual lift $X_{36,1+\epsilon}$ over $\mathbb{Z}_p[\epsilon]/\epsilon^2$ is smooth proper by §3, so its reduction is one of the first-order deformations covered by the lemma. Vanishing can be checked over $\overline{\mathbb{F}}_p$ and descends by faithful flatness. Thus its entire p-series lies in $p\mathbb{Z}_p[\epsilon]/\epsilon^2[[T]]$. Equations (II.5.1)–(II.5.6) now prove (II.1.5). This proves

$$W_n/n \in \mathbb{Z}_p \quad \text{for every inert } p > 5 \text{ and every } n \ge 1. \tag{II.7.5}$$

II.8 The singular prime 5

At 5 we use the singular fiber and its symmetries directly. No smooth first-order K3 statement is applied to it.

II.8.1 Three nodes and the central height

In characteristic 5, the target point $(s, t) = (36, 1)$ reduces to $(1, 1)$. By (II.3.4), the singular torus points of $X_{1,1}$ have all four coordinates equal to q, with $q^4 = (1 + q)^4 = 1$. Over the geometric field, $q^4 = 1$ already has all four roots in $\mathbb{F}_5$; excluding -1 leaves exactly

$$q = 1, 2, 3. \tag{II.8.1}$$

The boundary remains smooth by §3.1. At these torus points, eliminate $x_4 = 1/(XYZ)$. The Hessian of $\Lambda_1 - 1$, computed with Euler derivations in the three multiplicative local coordinates, is

$$\frac{q}{(1+q)^2}(I_3 + J_3), \qquad \det = 4\left(\frac{q}{(1+q)^2}\right)^3 \neq 0. \tag{II.8.2}$$

Here J_3 is the all-one matrix. Its determinants at the three points are respectively $1, 3, 3$ in $\mathbb{F}_5$. Since 2 is invertible, all three singularities are ordinary double points of type A_1.

Blow up these three points over $k = \overline{\mathbb{F}}_5$, obtaining $\rho : Y \to X_{1,1}$. A nondegenerate quadratic tangent cone is a smooth conic, so each blowup resolves its node with one exceptional $\mathbb{P}^1$ of square -2. The resolution is crepant: the ambient threefold blowup has discrepancy 2, canceled by the surface's multiplicity 2. The central complete intersection is Cohen–Macaulay and regular away from these isolated nodes, hence normal. It is connected by (II.3.2); a connected normal scheme here is integral. Normality gives $\rho_*\mathcal{O}_Y = \mathcal{O}_X$. The exceptional conormal is $\mathcal{O}_{\mathbb{P}^1}(2)$, whose successive powers have vanishing H^1. Formal functions gives $R^1\rho_*\mathcal{O}_Y = 0$; higher direct images vanish because the fibers have dimension at most one. Thus

$$R\rho_*\mathcal{O}_Y = \mathcal{O}_X. \tag{II.8.3}$$

Crepancy and (II.3.2) now show that Y is a smooth proper K3 surface.

The 24 boundary lines avoid the nodes. Their strict transforms retain the intersection matrix G indexed by all permutations c of the four roles:

$$G_{c,c} = -2, \qquad G_{c,d} = 1 \text{ if } d \text{ swaps the free role of } c \text{ with one fixed role},$$

with all remaining entries zero. Its exact characteristic polynomial is

$$\det(\lambda I - G) = \lambda^6(\lambda-1)(\lambda+1)^3(\lambda+2)^4(\lambda+3)^3(\lambda+4)^6(\lambda+5). \tag{II.8.4}$$

This specifies a finite integer determinant identity completely: there are precisely 24 permutations and precisely three off-diagonal entries 1 per row. Since G is symmetric over $\mathbb{R}$, its rank over $\mathbb{Q}$ is 18. The three exceptional curves are pairwise disjoint and disjoint from the boundary. Their addition gives the block intersection matrix $G \oplus (-2I_3)$, of rank 21. This is a rank statement for actual integer intersection numbers, not a reduction of the Gram matrix modulo 5. Hence $\rho(Y) \geq 21$.

The finite-height inequality $\rho + 2h \leq 22$ quoted in §6 forces Y to have height infinity. Its central formal Brauer five-series is therefore identically zero.

Equation (II.8.3) identifies this group with that of the singular $X_{1,1}$. Indeed, pull back the sheaves of principal nilpotent units and filter by powers of a nilpotent coefficient algebra J. Their graded pieces are structure sheaves tensored with vector spaces over k. Equation (II.8.3) identifies their cohomology in every degree. Induction through the finite filtration gives a natural isomorphism

$$H^2(X_{1,1}, 1 + \mathcal{O}_X \otimes_k J) \simeq H^2(Y, 1 + \mathcal{O}_Y \otimes_k J). \tag{II.8.5}$$

Consequently the singular central formal Brauer five-series is zero as well. Only constant nilpotent extensions of the central resolution have been used; a simultaneous resolution of the smoothing family is unnecessary.

II.8.2 The natural formal group of the singular two-parameter family

Work now over $R = \mathbb{Z}_5[[S, T_0]]$, with $s = 1 + S, t = 1 + T_0$. The complete intersection is flat, as in §3, even though its central fiber is singular. Section 5, including the ambient acyclicity (II.5.2), applies without a smoothness assumption. It supplies an integral one-dimensional formal group with logarithm (II.5.1), naturally isomorphic to the infinitesimal formal Brauer functor of this very family. Its central five-series is zero by (II.8.5).

The transformations E, I in (II.3.9) generate S_3: their coordinate Möbius transformations satisfy $E^2 = I^2 = (EI)^3 = 1$. They fix the central parameter point $(1, 1)$. By (II.3.10), their action on the dualizing residue there has factors $-1, -1$. Serre duality and (II.3.2) give the same sign character on the formal-group tangent $H^2(\mathcal{O}_X)$. This computation is valid for the singular Gorenstein complete intersection; it uses its dualizing form, not a smooth form obtained from a hypothetical simultaneous resolution.

On the parameter tangent space V, with vectors written (ds, dt), the matrices are

$$DE = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad DI = \begin{pmatrix} -1 & 0 \\ -1 & 1 \end{pmatrix}. \tag{II.8.6}$$

An invariant functional $a\,ds + b\,dt$ satisfies $a = b$ and $2a + b = 0$. Since 3 is a unit in characteristic 5, the only invariant functional is zero, also after any field extension.

II.8.3 Equivariance kills every first derivative of the five-series

We record the elementary formal-group argument. Suppose a one-dimensional formal group in characteristic $p > 2$ has zero central p-series. A group of family isomorphisms fixes the parameter point, acts on its formal-group tangent by a character χ valued in $\{1, -1\}$, and its parameter tangent representation V has no invariant covector. Then the entire first parameter derivative of its p-series is zero.

For a tangent vector v, write

$$[p]_{\epsilon v}(Z) = \epsilon h_v(Z), \qquad h_v(Z) = \sum_{m \geq 1} c_m(v) Z^m, \tag{II.8.7}$$

where $c_m \in V^*$. If some c_m is nonzero, choose the least such $m = n$, globally over the coefficient functionals. The endomorphism identity for the p-series implies

$$h_v(F_0(X, Y)) = h_v(X) + h_v(Y).$$

For v with $c_n(v) \neq 0$, its least-degree term gives $(X + Y)^n = X^n + Y^n$, so $n = p^r$. To see the last implication, write n in base p and use $(1 + X)^n = \prod_i (1 + X^{p^i})^{n_i}$: unless exactly one digit is 1 and the others are zero, there is an intermediate nonzero binomial coefficient.

For a family isomorphism $a_{g,v}$, naturality gives

$$a_{g,v}([p]_{\epsilon v}(Z)) = [p]_{\epsilon Dg(v)}(a_{g,v}(Z)).$$

The coefficient in degree ϵZ^n is $\chi(g)c_n(v) = \chi(g)^n c_n(Dg(v))$. Higher coordinate terms cannot contribute at the globally least degree; parameter derivatives of the isomorphism multiply the zero central series. Since $n = p^r$ is odd, $\chi(g)^n = \chi(g)$. Thus c_n is an invariant covector, a contradiction. This proves the claim for the entire first-order series, without choosing an additive coordinate on the family.

Apply the claim to the actual formal group of §8.2. Its central five-series is zero, both tangent character values are -1, and (II.8.6) has no invariant covector. Therefore

$$[5]_{X_{(1,1)+\epsilon v}} = 0 \quad \text{over } \overline{\mathbb{F}}_5[\epsilon]/\epsilon^2 \quad \text{for every } v. \tag{II.8.8}$$

It descends to $\mathbb{F}_5$ by faithful flatness.

II.8.4 Specialization to 36 and all orders of divisibility

Specialize the integral family by

$$S \mapsto 35, \qquad T_0 \mapsto \epsilon \quad \text{in } A = \mathbb{Z}_5[\epsilon]/\epsilon^2.$$

This continuous map from the completed parameter ring is well-defined: terms with T_0-degree at least two vanish, and the remaining series in powers of 35 converge 5-adically. Its reduction is the first-order direction $(ds, dt) = (0, \epsilon)$ at $(1, 1)$. Equation (II.8.8) says that the entire specialized five-series belongs to $5A[[Z]]$. The integral logarithm lemma of §5 and its exact constant-term comparison give (II.5.6) at $p = 5$, for every h. Thus (II.1.5) holds at 5, and (II.1.3) proves

$$W_n/n \in \mathbb{Z}_5 \qquad (n \geq 1). \tag{II.8.9}$$

II.9 Combination and finite certificates

Section 2 proved integrality at 2 and 3. Section 4 proved it at every $p > 5$ with $\left(\frac{-30}{p}\right) = 1$. Sections 6–7 proved it at every $p > 5$ with $\left(\frac{-30}{p}\right) = -1$; those are all remaining primes greater than 5. Section 8 proved it at 5. Consequently the rational number W_n/n is in $\mathbb{Z}_p$ for every prime p. A rational number integral at every prime is an integer. This proves (II.0.1) for every positive integer n, including $n = 1$, when $W_1 = 7$.

The finite exact assertions used above are (II.3.11), (II.6.3), and (II.8.4), together with the three Hessians and tangent matrices at 5. Each matrix is completely specified in the text. The linked executable `compiled_integrality_certificates_v1.py` independently constructs the 24 boundary-role graph, its toric divisor sums and the nine anti-invariant differences; it checks the two displayed 9-by-9 matrices, their determinants, and the characteristic polynomial (II.8.4). It also verifies the five-adic node list, Hessians, and parameter representation. These finite calculations certify explicit finite identities used in the geometric proof. No bounded numerical testing is used to infer an unbounded divisibility statement. The reductions from those identities to all primes, all exponents and all cofactors are the arguments above.

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